Exact Avoid-or-Compress Extrema for Dense Reachable Images: Defects Four and Five

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Abstract

Let S be a reachable image of a strongly connected synchronizing automaton and $\varnothing \neq A \subsetneq S$. We determine the maximum shortest avoid-or-compress length over binary and arbitrary finite alphabets for $m = n - |S| = 4$ and $m = 5$. The exact values are $\min(n - |A|, 11)$ and $\min(n - |A|, 14)$, respectively. Uniform binary constructions attain both bounds. For the upper bounds, short-pair graphs and actual-image constraints reduce all parameters to finite necessary domains. Exhaustive checks exclude the domains, apart from a single relaxation eliminated by a multipartite overlap lemma. The largest calculation covers 3,206,988 domains.

1 Problem and results

All words act on the right and length counts *original alphabet letters*. A complete deterministic automaton is a triple $\mathcal{A} = (Q, \Sigma, \delta)$, where Q and Σ are finite and δ is total. It is strongly connected if every state reaches every other state, and synchronizing if $|Qw| = 1$ for some word w. Let $n = |Q|$, let $S = Qv$ be an actual reachable image with $|S| = s$, and let $\varnothing \neq A \subsetneq S$, $|A| = r$. The admissible parameters are $n \geq 2$ and $1 \leq r < s \leq n$. Put

$$\mathrm{ac}(S, A) = \min\{|w| : A \not\subseteq Sw \text{ or } |Sw| < s\}, \qquad c(S) = \min\{|w| : |Sw| < s\}.$$

Thus avoidance means omitting *at least one* protected state, and $\mathrm{ac}(S, A) \leq c(S)$. Let $\mathcal{F}(n, s, r)$ maximize $\mathrm{ac}(S, A)$ over strongly connected synchronizing automata with arbitrary finite alphabets; $\mathcal{F}_{\mathrm{bin}}$ restricts to binary alphabets. Write $m = n - s$ and $q = n - r$. The avoid-or-compress lemma [6, 7] and the Frankl–Pin compression bound [2, 5] give

$$\mathcal{F}_{\mathrm{bin}}(n, s, r) \leq \mathcal{F}(n, s, r) \leq \min\{q, \tbinom{m+2}{2}\}. \tag{1}$$

A proof of the q bound appears in Appendix C. Open Problem 4 of the survey [7] asks for tight bounds depending on the set cardinalities. The two-state image case is already illustrated there, and related fixed-state avoiding-or-compressing sharpness is discussed in [1]. Our main contribution concerns entire dense-image parameter families.

Theorem 1 (Dense-image extrema; computer-assisted upper bounds)**.** *For all admissible parameters,*

$$\boxed{\mathcal{F}_{\mathrm{bin}}(n, n-4, r) = \mathcal{F}(n, n-4, r) = \min(q, 11)}$$

and

$$\boxed{\mathcal{F}_{\mathrm{bin}}(n, n-5, r) = \mathcal{F}(n, n-5, r) = \min(q, 14).}$$

In every strongly connected synchronizing automaton, every $(n-4)$-subset has $c(S) \leq 11$ when $n \geq 11$, and every $(n-5)$-subset has $c(S) \leq 14$ when $n \geq 16$. These pure-compression assertions do not require reachability.

The lower bounds are established by explicit binary automata. For the upper bounds, Sections 3 and 4 present the short-pair reductions, exhaustive finite-domain coverage, and computational basis. Appendix A develops the structural and finite-domain lemmas supporting both upper bounds. The uniform q bound supplies the remaining parameter ranges. The codimension-three input $C_3 \leq 9$ is recorded in Lemma 9. The same two extrema hold if S is allowed to be an arbitrary $(n-m)$-subset: neither the universal q bound nor the upper proofs requires reachability, while all matching lower witnesses are reachable.

Proposition 2 (Binary sharpness below the two-block threshold)**.** *Let $m \geq 4$ and $m+1 \leq q \leq 2m+1$. For every admissible r there is a binary strongly connected synchronizing automaton with a reachable $(n-m)$-image S and an r-set $A \subsetneq S$ such that $\mathrm{ac}(S, A) = q$.*

Proof. Appendix C supplies all three required constructions, including their reaching and reset words. The first cardinality face covers $q = m+1$; the second covers $q = m+2$. For $m+3 \leq q \leq 2m+1$, take $g = q-m-1$ and $M = q-1$ in Proposition 29. Then $g \geq 2$, $M \geq 2g$, $n = r+M+1$, $s = r+g+1 = n-m$, and $\mathrm{ac}(S, A) = M+1 = q$. □

2 The binary witness

Figure 1 gives the two original-letter actions for a fourteen-state instance, including the initial set and unique merger. The potential and cell arguments below, not the displayed attaining words alone, establish the lower bound for every word.

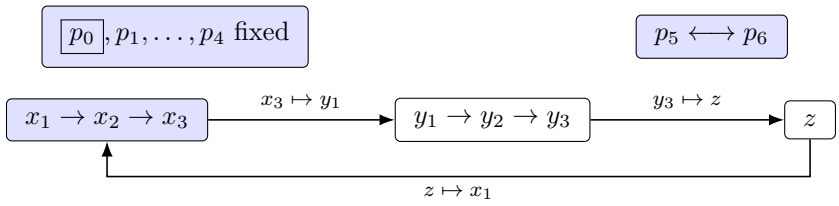
2.1 Uniform construction and proof

Theorem 3. *Put $m = n-s \geq 3$ and $q = n-r \geq 2m+2$. There is a binary strongly connected synchronizing automaton with reachable S attaining $\mathrm{ac}(S, A) = \min\{q, 3m-1\}$. Consequently $\mathcal{F}_{\mathrm{bin}} = \mathcal{F} = q$ whenever $2m+2 \leq q \leq 3m-1$. For larger q the construction gives only a lower bound.*

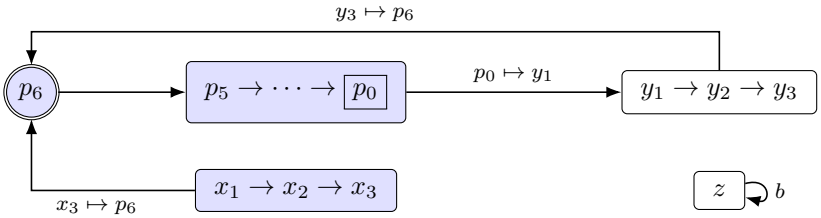
Proof. Write $t = m-1 \geq 2$ and $p = n-2t-1 \geq r+3$. Take $P = \{p_0, \ldots, p_{p-1}\}$, $X = \{x_1, \ldots, x_t\}$, $Y = \{y_1, \ldots, y_t\}$ and one further state z. Letter a interchanges the last two P-states, fixes the others, and cycles $x_1, \ldots, x_t, y_1, \ldots, y_t, z$. Letter b sends p_0 to y_1, p_i to p_{i-1} for $i \geq 1$, increments indices in X, Y below t, sends both x_t, y_t to p_{p-1}, and fixes z. Put $S = P \cup X$, $A = \{p_0, \ldots, p_{r-1}\}$ and $L = 3t+2$.

No compression before L. For $\rho_i = \min\{i, p-2\}$ define a symmetric potential Φ on unordered pairs, zero on singleton pairs, by the following table. Indices in X, Y run from 1 to t, and indices in P from 0 to $p-1$.

(a) Letter a



(b) Letter b



Blue: initially occupied $S = P \cup X$. Boxed p_0: protected. Double circle: merger target.

Figure 1: The two original-letter actions for $t = 3$, $p = 7$, $r = 1$ ($n = 14$). Both incoming arrows at p_6 are shown: the only nontrivial kernel of b is $\{x_3, y_3\} \mapsto p_6$. Letter a is a permutation. The words a^6ba^3b and a^6bab^5 attain compression in 11 letters and avoidance of p_0 in 13 letters, respectively; optimality follows from the all-word arguments below.

pair	potential
P–P, P–X, X–X (distinct)	L
$\{x_i, y_j\}$, $j = i$	$t - i + 1$
$\{x_i, y_j\}$, $j > i$	$2t - i + 1$
$\{x_i, y_j\}$, $j < i$	$L - j$
$\{x_i, z\}$	$2t - i + 1$
$\{y_i, y_j\}$, $i < j$	$L - j$
$\{y_i, z\}$	$2t + 1$
$\{p_i, y_j\}$	$\min\{L, \max\{L - j, 2t + 3 + \rho_i\}\}$
$\{p_i, z\}$	$\min\{L, 2t + 2 + \rho_i\}$

Substitution in the two letter actions gives

$$\Phi(E) \le 1 + \Phi(Ea), \qquad \Phi(E) \le 1 + \Phi(Eb).$$

Here are the boundary checks. The last-two-P interchange preserves ρ; a downward P-step decreases it by at most one. For P–Y pairs the a-boundary $y_t \to z$ decreases the second term from $2t+3+\rho$ to $2t+2+\rho$. The b-boundary $p_0 \to y_1$ with $j < t$ has source $L - j$ and target $L - j - 1$, since $j \le t - 1$; with $j = t$ its target is at least $L - 1$. For X–Y, crossing x_t under a gives Y–Y with potential $L - j - 1$ when $j < i$; crossing under b gives P–Y of at least that potential. The kernel pair $\{x_t, y_t\}$ has potential one. Crossing y_t in a Y–Y pair under a gives Y–z of potential $2t + 1 = L - t - 1$. For X–z, the a-boundary is $\{x_t, z\} \to \{y_1, x_1\}$, with values $t + 1, t$; for Y–z, every a-image has potential $2t$. All remaining cases either retain potential, decrement an index by one, or enter a type whose potential is at least the source minus one. Clipping at L cannot increase a positive difference. Thus the inequalities hold for every index, not just tested sizes. Every distinct pair in S has potential L. Telescoping along a merging word proves its length is at least L; every compression merges an initial pair, giving the claimed lower bound.

No avoidance before q. Use singleton P-cells and t color cells, giving x_i, y_i color i and z color 1.

Let T_a swap the last two P-cells and cycle the colors, and let T_b cycle $p_0, 1, \dots, t, p_{p-1}, \dots, p_1$. For the incidence matrix C both letters satisfy

$$D_\ell C = CT_\ell + f_z(e_1 - e_2).$$

Here D_ℓ is the original-letter transition matrix, f_z the z-column, and e_i the color row; the displayed discrepancy is a rank-one outer product. Since S has one state in every cell, induction along each original word w gives $\chi_S D_w C = \mathbf{1} + \sum_j \mu_j(e_1 - e_2)T_{w_j}$, where $\mu_j \geq 0$ counts trajectories at z immediately before the corresponding suffix w_j. A trajectory from X needs at least $t+1$ steps to reach z; any entrance from P starts at y_1 after at least one step and then needs at least t more. Thus the first nonzero correction has position at least $t+2$. In the prescribed cell graph, color 2 needs $t-1$ letters to enter P at p_{p-1} and at least $p-r$ more to reach A. The only interchange is between adjacent last P-points and cannot shorten this descending path. A negative correction therefore reaches A only at length at least $(t+2)+(t-1+p-r) = p+2t+1-r = q$. Before then every protected coordinate has multiplicity at least one.

Attainment. After $a^{2t}b$, the missing P-point is p_{p-1}; after a further a it is p_{p-2}. This is the unique b^{p-r-1}-preimage of p_{r-1}. Hence $a^{2t}bab^{p-r-1}$ omits that protected point at length q. The $X \cup Y \cup \{z\}$ portion after $a^{2t}b$ is $\{x_2, \dots, x_t, y_1, z\}$. After a^t it contains x_t and y_t, so $a^{2t}ba^tb$ compresses at length L. The existence of x_t in the preceding set uses $t \geq 2$. Combining both upper witnesses with the lower bounds gives $\mathrm{ac}(S, A) = \min\{q, L\}$.

Structural requirements. Put $B = P \cup Y$, $h = p+t$ and $E = B \cup \{z\}$. Letter b cycles B in the order $p_0, y_1, \dots, y_t, p_{p-1}, \dots, p_1$, and $Qb^t = E$. The a-cycle connects all other states to B, proving strong connectivity. On E, $d = a^{2t}b^h$ fixes P, cycles $y_1 \to y_t \to \cdots \to y_1$, and sends z to y_t. Thus $e = d^t$ fixes B and sends z to y_1. Since $Ba^{t+1} = S$, the original-letter word b^tea^{t+1} reaches S. The word a^{2t+1} fixes $X \cup Y \cup \{z\}$ and transposes the last two P-points, adjacent on the b-cycle. These words generate every permutation of B, with inverses realized by positive powers. Moreover $Ba = (B \setminus \{y_1\}) \cup \{z\}$, $Bab = (B \setminus \{y_2\}) \cup \{z\}$, and $Babe = B \setminus \{y_2\}$. Permutations route any pair in a current image into the kernel of abe; repeated compression synchronizes B. Prepending b^te synchronizes Q. For $q \leq L$, the construction and the universal bound $\mathrm{ac}(S, A) \leq q$ determine both maxima. □

3 Short-pair graphs and the upper bounds

Let P_h join distinct states if some original-letter word of length at most h merges them. Write $\tau(P_h)$ for its minimum vertex-cover size. A set survives compression through depth h exactly when it is independent in P_h. Thus

$$\exists T,\ |T| = n - m,\ c(T) > h \quad \Longleftrightarrow \quad \tau(P_h) \leq m. \tag{2}$$

This is the short-pair viewpoint on subset compression; see also the computational compression framework in [4]. Let $K_h = \{v \in Q : \deg_{P_h}(v) > 0\}$ be the nonisolated core and $J_h = Q \setminus K_h$ the isolated-state reservoir. If T is an independent $(n-m)$-set, its complement $C = Q \setminus T$ is a vertex cover of size m. As isolated vertices cover no edges, $C \cap K_h$ is also a vertex cover. Consequently

$$|(Q \setminus T) \cap J_h| \leq m - \tau(P_h). \tag{3}$$

In particular, when $\tau(P_h) = m$, every such T contains all of J_h:

$$J_h \subseteq T, \qquad Q \setminus T \subseteq K_h.$$

Thus, independently of how large n is, all m omitted states lie in the nonisolated core. Figure 2 depicts this finite named-state reduction, including the exceptional isolates in the lower-cover branches.

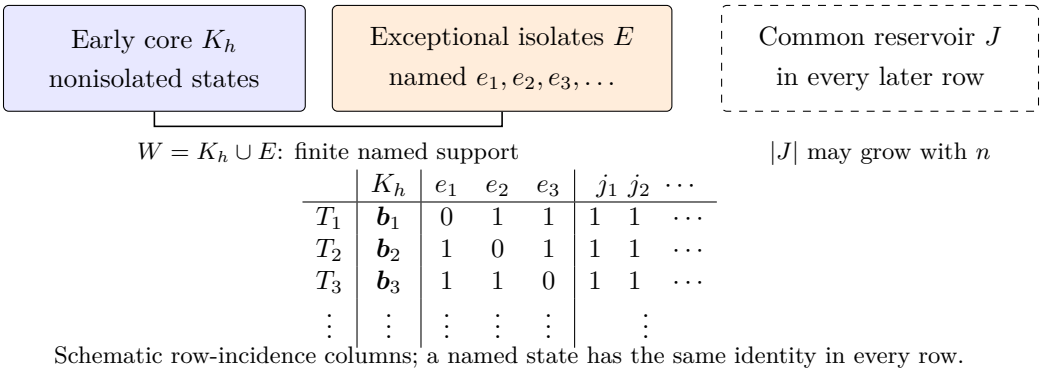


Figure 2: The finite reduction retains more than the early graph. For the complete maximum-independent-set family $\mathcal{D}$ of P_H, set $E = \bigcup_{T\in\mathcal{D}}(J_h \setminus T)$ and $J = J_h \setminus E$. The displayed pattern illustrates the at-most-one exceptional omission per row when $\tau(P_h) = m-1$. The support reductions give $|W| \le 10$ for $(m, h, H) = (4, 7, 9)$ and $|W| \le 11$ for $(5, 9, 11)$ in their lower-cover branches. Identical membership columns do not merge automaton states: finite letter signatures must retain reservoir cardinality constraints and consistent same-letter identities, including in square tests.

If $\tau(P_h) = m - 1$, equation (3) allows at most one omitted isolated state in each row. The lower-cover branches retain these exceptional omissions by name, alongside the nonisolated early core; only the common, never-omitted isolated states form the unbounded reservoir. The short-image, support-deficit and row-rank reductions in Appendix A bound the named support independently of n.

Lemma 4 (Safety and union of actual short images)**.** *Suppose $|S| = n - m \ge 2$, $c(S) \ge L$, and $0 \le d \le L - 2$. Every Sw with $|w| \le d$ has size $n - m$ and is independent in P_{L-1-d}. In a strongly connected automaton their union has at least* $\min(n, n - m + d)$ *states.*

Proof. Compression in the prefix w, or a merging suffix of length at most $L - 1 - d$, would compress S before length L. Let $U_i = \bigcup_{|w|\le i} Sw$. If U_i is proper, strong connectivity gives an edge from U_i to its complement, whence $|U_{i+1}| \ge |U_i| + 1$. Starting at $|U_0| = n - m$ proves the union bound. □

For an early layer $G = P_h$ and a later layer P_H, the reductions retain the *complete* later maximum-independent-set family $\mathcal{D}$. Every nonisolated early-core state and every early isolated state omitted by some row is named consistently across all rows. The other isolated states lie in the common reservoir. Finite letter signatures record their possible entrances and exits. The support bounds and row-rank restrictions of Lemmas 15 and 19 reduce arbitrary n to finitely many named domains.

Every original letter projects to an admissible signature. The finite domains enforce the one-step pair-distance inequalities, the existence of attaining letters, compatible row exits, and necessary strong connectivity. Where $H - h = 2$, the same original letter must also satisfy the safe-row condition for its square f^2. These constraints hold for every automaton satisfying the proposed long-compression hypothesis, so excluding all compatible finite domains proves the corresponding upper bound.

The upper-bound exclusions use the short-pair cover number to split the finite search. Table 1 summarizes the complete branch coverage.

Hypothesis	Branch	Finite scope	Exclusion
$m = 4$, $c(S) \geq 12$	$\tau(P_7) = 4$	91 cores	finite checks
$m = 4$, $c(S) \geq 12$	$\tau(P_7) = 3$	10,142 named domains	finite checks
$m = 5$, $c(S) \geq 15$	$\tau(P_9) = 5$	2,294 cores	finite checks
$m = 5$, $c(S) \geq 15$	$\tau(P_9) = 4$	$3{,}595 + 3{,}206{,}988$ $= 3{,}210{,}583$ domains	finite checks plus Lemma 5 for the lone historical relaxation

Table 1: Coverage for $|S| = n - m$, with $n \geq 11$ for $m = 4$ and $n \geq 16$ for $m = 5$. The last row separates eleven-point early support from early support five to ten; the latter count is after 38,398 static exclusions. Each row records a complete exclusion of necessary domains derived from arbitrary finite-alphabet automata. The resulting bounds are $c(S) \leq 11$ and $c(S) \leq 14$, respectively.

3.1 Defect four: excluding twelve steps

Assume $n \geq 11$, $|S| = n - 4$, and $c(S) \geq 12$. The known codimension-two bound $C_2 \leq 6$ and codimension-three bound $C_3 \leq 9$ imply $\tau(P_7) \geq 3$ and $\tau(P_9) \geq 4$, whereas (2) gives $\tau(P_7) \leq 4$. Here C_j denotes the maximum compression distance over all $(n - j)$-sets in the same automaton. There are exactly two branches.

1. $\tau(P_7) = 4$. The actual images of depth at most four are maximum independent sets of P_7 and cover Q by Lemma 4. Lemma 20 bounds the nonisolated core size by $5 \leq N \leq 8$, its maximum degree by four, and its edge count by $7 \leq e \leq 10$, with maximum-family row rank at least five. The complete classification has 91 isomorphism classes. The complete letter and distance checks leave 152 joint distance vectors, all excluded by the necessary exit from a proper core.

2. $\tau(P_7) = 3$. Then $\tau(P_9) = 4$. The Lemma 15 gives 117 early graph classes with $5 \leq N \leq 9$ and at most ten named states in total. The complete later family has three to seven rows, row rank at least three, and union omitting at most two states. The closure-compatible families, modulo core automorphisms and consistent isolated-state renaming, give exactly 10,142 necessary domains. Every domain is excluded by the common distance, quotient connectivity, and, where needed, same-letter square conditions.

The proper-core conditions apply because $n \geq 11$ exceeds the named support. The six disjoint computational ranges contain $1{,}414 + 259 + 978 + 2{,}349 + 2{,}500 + 2{,}642 = 10{,}142$ domains, with no survivors. Both complete letter-generation orders and both common-distance implementations were compared in these ranges. Together, the two branches prove $c(S) \leq 11$. For $n \leq 10$, $q \leq 9$ suffices. For all n, matching binary witnesses come from Proposition 2 at $q = 5, \ldots, 9$ and Theorem 3 at $q \geq 10$.

3.2 Defect five: excluding fifteen steps

Assume $n \geq 16$, $|S| = n - 5$, and $c(S) \geq 15$. Since $C_3 \leq 9$, (2) gives $\tau(P_9) \in \{4, 5\}$. Lemma 21 gives the following exhaustive split.

1. $\tau(P_9) = 5$. The actual images of depth at most five are maximum independent sets and cover Q. The core has $6 \leq N \leq 10$, maximum degree at most five, at least nine edges, and maximum-family row rank at least six. Exactly 2,294 classes survive the complete graph classification. The necessary original-letter defect is at most three. Indeed, Lemma 8, with $m = 5$, word length $a = 1$, and defect $d = 4$, gives $c(S) \leq 2^{5-4}(1 + 4 + 2) - 1 = 13 < 15$; larger

defects give still shorter bounds. The row-exit threshold is six and fifth powers remain safe. The joint distance calculation excludes all 2,294 classes. Its 72 disjoint ranges are $[32j, 32j+31]$ for $0 \le j \le 70$, and $[2272, 2293]$, so $71 \cdot 32 + 22 = 2294$.

2. $\tau(P_9) = 4$. Take $\mathcal{D}$ to be the complete maximum-independent-set family of P_{11}. Here $C_4 \le 11$ excludes an independent $(n-4)$-set, while S is independent, so $\alpha(P_{11}) = n-5$. The family has row rank at least four and union omitting at most two states. The named support is at most eleven. The complete early-core classification has 4,658 classes: 90 with eleven nonisolated states and 4,568 with five to ten. Each later row is either an early maximum set with one point deleted, or a nonextendable near-maximum set; each early maximum set contributes at most two later rows. The finite closure enumeration retains the identities of all omitted early isolated states.

At eleven-point early support Lemma 22 forces $\tau(P_{10}) = 4$ and exactly two early maximum sets R, T, differing in one state. Exactly one is independent in P_{10}. That same choice $R_0 \in \{R, T\}$ must be preserved into $\{R, T\}$ by *every* original letter; allowing a separate choice for each letter would discard an essential constraint. The eleven-point coverage in Table 1 of the 3,595 domains is

$$\underbrace{883 + (1201 - 19)}_{\text{two omitted states: } 2065} + \underbrace{1504 + 26}_{\text{one omitted state: } 1530} = 3595.$$

The 19-domain overlap is counted once. The 1,504 single-omission domains and the final 26-domain component were checked, including both common R_0 choices in the latter.

For early support five to ten, two cores are first excluded by Lemma 23. The remaining closure enumeration and checks give

$$\underbrace{3{,}245{,}386}_{\text{raw domains}} - \underbrace{38{,}398}_{\text{static exclusions}} = \underbrace{3{,}206{,}988}_{\text{completed dynamic checks}}.$$

These counts use the original joint distance predicate. Under the original predicate, exactly 3,206,987 dynamic domains were unsatisfiable. The sole satisfying *relaxation* was core 447, domain 0: the ninth-layer core is $K_{4,4}$ with parts $\{0, 1, 2, 3\}$ and $\{4, 5, 6, 7\}$, and the four later rows on that core are

$$\{0, 1, 2\}, \quad \{0, 1, 3\}, \quad \{4, 5, 6\}, \quad \{4, 5, 7\}.$$

All rows also contain the common isolated states. The current joint predicate incorporates Lemma 24 and excludes this domain directly. The verification specification in Section 4 records both predicates and their corresponding outputs.

The following paper argument excludes this remaining domain.

Lemma 5 (Multipartite overlap exclusion)**.** *Let $h \ge 1$ and $t \ge 2$. Suppose a noncomplete P_h in a synchronizing automaton is a complete multipartite graph on $V = V_1 \sqcup \cdots \sqcup V_t$ and has isolated remaining states J. Assume each V_i admits a family $\mathcal{R}_i$ of nonempty subsets whose union is V_i, whose intersection graph is connected, and for which every $R \cup J$ with $R \in \mathcal{R}_i$ is independent in P_{h+1}. Then this configuration is impossible.*

Proof. A P_h nonedge cannot be merged in one letter. Fix a letter f. It is therefore injective on each part and on J, and Jf is disjoint from Vf. Since every $(R \cup J)f$ is P_h-independent, if $Jf \subseteq J$, then $Jf = J$ and Rf is contained in some single part. Intersecting R, R' have intersecting images, so the connected intersection graph forces all of $V_i f$ into one part. Every original nonedge remains a nonedge.

Otherwise Jf meets some part V_j. Independence of $(R \cup J)f$ forces $Rf \subseteq V_j \cup J$ for every R, hence $Vf \subseteq V_j \cup J$ and $Jf \subseteq V_j \cup J$. Again every original nonedge remains a nonedge. This holds for every letter. But a synchronizing automaton with noncomplete P_h must have some letter sending a nonedge to an edge of P_h: follow the shortest merging word for a pair at distance greater than h until the distance first drops from $h+1$ to h. Contradiction. □

The two pairs of intersecting triples above separately cover the two parts of $K_{4,4}$, and their rows (with common isolates adjoined) are independent in P_{11}, hence in P_{10}. Lemma 5 with $h = 9$ excludes the only relaxation survivor. Thus the combined complete finite exclusions establish $c(S) \leq 14$. For $n \leq 15$, the original bound $q \leq 14$ suffices. Matching binary witnesses are given by Theorem 3 for $q \geq 12$ and Proposition 2: the first and second cardinality faces cover $q = 6, 7$, and the cofinal construction covers $q = 8, \ldots, 11$.

4 Computational basis and reproducibility

The upper proof combines the short-pair and finite-support lemmas of Appendix A, the complete finite-domain checks specified below, and the overlap exclusion in Lemma 5. The version-pinned implementation and exact reproduction commands are documented in [8]. The principal source files are:

Component	Principal source files under `scripts/problem4/`
Defect four, cover four	`compression_fourteen_cores.cpp`
Defect four, cover three	`low_cover_core_domains.cpp`; `low_cover_actions.cpp`
Defect five, cover five	`rank_core_graph6.cpp`; `near_max_rank_cores.cpp`; `near_max_rank_actions.cpp`
Defect five, cover four	`fifteen_low_cover_domains.cpp`
Joint necessary constraints	`pair_sat_control.hpp`
Binary construction controls	`check_two_block.py`; `two_block_potential.py`

The largest historical finite calculation covered 3,206,988 necessary domains using 24 workers, paired letter-generation orders, and CaDiCaL 3.0.1 with joint SAT constraints (22,923.697 seconds). Of these, 3,206,987 were unsatisfiable under the checked relaxation. The remaining model is eliminated by Lemma 5. The original joint predicate applies distance and row constraints. The current implementation strengthens it with the proper-layer entrance condition of Lemma 24. The option `--historical-no-pair-entrance`, combined with `--distances` and `--joint-sat`, selects the original predicate. For core 447, domain 0, both modes produce 1,768 letter signatures: the historical mode gives `joint=1`, whereas the current default gives `joint=0`. The indicators record satisfiability of necessary relaxations. Every UNSAT result under the original predicate also holds for the strengthened predicate. Lemma 5 supplies the paper exclusion of the original predicate's single survivor.

For the cover-five branch, the graph6 driver is `rank_core_graph6.cpp`, with parameters `5 6 9 --actions START COUNT THREADS --sat-distances`. Its input is the complete union of the nauty/Traces `geng` 2.9.3 streams for orders $6, \ldots, 10$, each generated with `-d1 -D5`; no connectedness restriction is imposed. The ranges stated in Section 3.2 cover all 2,294 retained cores.

Exact commands, including the single-domain comparison of the two predicates, are given in `docs/RUNNING.md` in the source revision linked above. The complete calculation covers the 3,206,988 domains in Table 1; the selected-domain comparison fixes the two predicate versions at core 447, domain 0. The verification record consists of the canonical domain coverage, terminal enumeration

summaries, solver results and the versioned source. Cross-checks compare both generation orders, exact small-domain enumerations and literal automaton actions. The reproducibility contract in Appendix B specifies the coverage and result checks.

5 A conjectured extension

The binary lower-bound constructions give $\mathcal{F}_{\mathrm{bin}}(n, n-m, r) \geq \min(q, 3m-1)$ for every $m \geq 4$, and Theorem 1 establishes equality for $m = 4, 5$.

Conjecture 6 (Linear defect formula)**.** *For every $m \geq 4$ and all admissible n, r,*

$$\mathcal{F}_{\mathrm{bin}}(n, n-m, r) = \mathcal{F}(n, n-m, r) = \min(n-r, 3m-1).$$

The remaining parameter region for exact avoid-or-compress extrema is $m \geq 6$ and $q \geq 3m$.

A Structural reductions and finite verification

This appendix specifies the mathematical implications that connect arbitrary automata with the finite domains of Section 4. All maps and powers below act in the original alphabet. For a word u, write $\mathrm{rk}(u) = |Qu|$ and $d(u) = n - \mathrm{rk}(u)$.

Lemma 7 (Rank transfer)**.** *Let u have length a and rank $k \geq 2$ in a synchronizing automaton. Every k-set T satisfies*

$$c(T) \leq a + n - k + 1.$$

Proof. Let $K_1, \ldots, K_k$ be the kernel blocks of u, and put $d = n - k + 1$. Assume all words of length at most $a + d$ preserve the cardinality of T. For every $|v| \leq d$, the k trajectories of Tv visit the k blocks once each; otherwise vu compresses T within $a + d$ letters. Consequently the characteristic row of Tv lies in the linear subspace

$$H = \{x \in \mathbb{R}^Q : \sum_{q \in K_1} x_q = \cdots = \sum_{q \in K_k} x_q\}, \qquad \dim H = n - k + 1 = d.$$

Let $V_i = \mathrm{span}\{\chi_T D_v : |v| \leq i\}$. A plateau $V_i = V_{i+1}$ makes V_i invariant under every letter. It then contains the row obtained from a reset word, which has mass k concentrated in a single kernel block; that row lies outside H. Hence $\dim V_0 < \dim V_1 < \cdots < \dim V_d$, yielding $\dim V_d \geq d + 1$, a contradiction. □

Lemma 8 (Iterated rank descent)**.** *For $m \geq 1$, $|S| = n - m \geq 2$, and an original word u of length a and defect $d = d(u) \geq 1$,*

$$c(S) \leq \begin{cases} a, & d > m, \\ 2^{m-d}(a + d + 2) - 1, & 1 \leq d \leq m. \end{cases} \tag{RD1}$$

Proof. The case $d > m$ follows from the cardinality of Qu. When $d = m$, Lemma 7 applies to the $(n-m)$-set S and gives $c(S) \leq a + m + 1$. For $d < m$, the proof of Lemma 7, applied to $T = Qu$, supplies a prefix v of length at most $d + 1$ for which vu compresses Qu. Thus $Q(uvu)$ has rank at most $n - d - 1$. The word uvu has length at most $2a + d + 1$. Induction on $m - d$, together with monotonicity of the right-hand expression in the attained defect, gives

$$c(S) \leq 2^{m-d-1}(2a + 2d + 4) - 1 = 2^{m-d}(a + d + 2) - 1.$$

Every concatenation counts its full original-letter length. □

Lemma 9 (Codimension-three input)**.** *In a strongly connected synchronizing automaton with $n \geq 6$, every $(n-3)$-set has a compressing word of length at most nine.*

Proof. The complete nine-case argument is provided below. All lengths refer to original letters, and all hole sets are complements of actual images. □

A.1 The codimension-three input

Tight layers, actual images and rank

Assume $c(S) = 10$, which is the only remaining case under the Frankl–Pin bound. The refinement of [4] gives $c(S) \leq 10 - p(P_h) + h$, where $p(P_h)$ is the maximum length of an $(n-3)$-set Frankl–Pin sequence using pairs from P_h. Any $q \leq 4$ distinct pairs admit a sequence of length q: for its ith pair choose three holes outside that pair, including an endpoint outside it of each later pair, and fill the remaining holes arbitrarily. There are at most three required endpoints and $n - 2 \geq 4$ available states. The pair is in its own subset and absent from every earlier subset. Consequently $|P_h| \leq h$ for $h = 1, 2, 3$. Pair layers grow strictly until complete, so $|P_h| = h$ at these three depths. In particular every letter is a permutation or has defect one with the common kernel 01.

Write $R_i = \{Sw : |w| \leq i\}$ and $V_i = \operatorname{span}\{\chi_S D_w : |w| \leq i\}$. Until compression $R_i \subsetneq R_{i+1}$: a plateau is forward invariant and contradicts synchronization. Also $\dim V_i \geq \min(i+1, n)$. Indeed a plateau in the vector spaces is invariant under all letters; a reset word followed by a path to any state yields all rows $|S|e_q$. Before compression these trajectory rows equal the characteristic rows of the actual images. Every image through depth six is independent in P_3. If $\mathcal{H}$ is its three-hole cover family and tests have maximum length L, the domain preserving $\mathcal{H}$ under these tests contains R_{6-L}. Such a domain therefore has at least $7 - L$ rows. Exiting it permits compression with at most $L + 3$ additional letters. For a permutation f, the hole action is $H \mapsto Hf$. For a defect-one letter with missing output e, it is $H \mapsto \{e\} \cup Hf$ when $01 \subseteq H$, and $H \mapsto \{e\} \cup (H \setminus \{0, 1\})f$ otherwise, for the safe covers under discussion. Thus the case tables below are direct set substitutions.

Common low-rank and five-cycle exclusions

For the low-rank cases the following table lists the *occupied core sets*; a common true outside set J is adjoined to every row.

case	occupied core rows	contains
triangle C; type 4 closed	$24, 23, 14, 13, 04, 03$	R_5
type 4 mixing	$145, 135, 045, 035, 124, 123$	R_5
type 7 D6closed	$14, 04, 13, 03, 12, 02$	R_5
type 7 D6Tmix	$135, 035, 125, 025, 134, 034$	R_5
type 7 D6Amix	$145, 045, 125, 025, 134, 034$	R_5
type 7 C_A	$34, 13, 03, 12, 02$	R_4
type 7 C_V	$14, 13, 03, 12, 02$	R_4
type 7 D_w	$135, 035, 125, 025, 134$	R_4

In the first five rows, characteristic rows X_i in the displayed order satisfy $X_1 - X_2 = X_3 - X_4 = X_5 - X_6$; these are two independent relations. For C_A, C_V, $X_2 - X_3 = X_4 - X_5$; for D_w, $X_1 - X_2 = X_3 - X_4$. Every rank is at most four. All coefficients sum to zero, so adjoining J preserves the relations. This contradicts $\dim V_4 \geq 5$.

The following occupied-core families each form a five-cycle on five states:

type 2 mixing B / closed T	$14, 04, 23, 12, 03$
type 3 mixing T	$24, 04, 13, 03, 12$
type 6 closed II	$34, 14, 12, 03, 02$
type 7 C_T	$34, 14, 03, 12, 02$

Each contains R_4, with the common set J adjoined. Since every letter has defect at most one, $\dim(V_3 D_f) \geq 3$. If a point of J entered a core point z, every row of $V_3 D_f$ would be spanned by domain rows containing z; there are only two, a contradiction. Thus every letter preserves the nonempty proper J (nonempty because $n \geq 6$), contradicting strong connectivity. The classifications below identify exactly where these families arise; the remaining cases use the displayed direct counts.

Write $E_1 = \{0, 1\}$. Up to simultaneous state relabeling the ordered three pairs, for $n \geq 6$, have precisely these nine types:

type	E_2	E_3
0	{0,2}	{1,2}
1	{0,2}	{0,3}
2	{0,2}	{1,3}
3	{0,2}	{2,3}
4	{0,2}	{3,4}
5	{2,3}	{0,2}
6	{2,3}	{0,4}
7	{2,3}	{2,4}
8	{2,3}	{4,5}

Proof of completeness: E_2 shares one endpoint with E_1 or is disjoint. In the first case normalize it to {0,2}; the ordered pair stabilizer fixes 0,1,2, and E_3 can be the remaining old edge, an edge from each of those three vertices to one new vertex, or an edge on two new vertices. In the disjoint case normalize E_2 to {2,3}. Its stabilizer independently swaps the endpoints of the first two pairs. Thus E_3 joins the pairs, joins either pair to a new point, or uses two new points. These give the five and four types above. No missing-state or alphabet-size assumption is used in this classification.

Type 1 is incompatible with strong connectivity Suppose $P_1 = 01$, $P_2 = \{01, 02\}$, $P_3 = \{01, 02, 03\}$. For every letter f, preimages of 01 are in P_2, and preimages of 02 are in P_3.

If f is a permutation, $f\hat{}\{\text{-}1\}(01)$ is 01 or 02, and $f\hat{}\{\text{-}1\}(02)$ is a different edge of the three-edge star. Their intersection is {0}. Since 01 and 02 also intersect at {0}, bijectivity gives $f\hat{}\{\text{-}1\}(0)=0$. In particular no other state maps to 0 under f.

If f is singular, let d be its unique missing output and c the common image of 0,1. If d is not in 01 and c is in 01, the preimages of 01 are the two distinct pairs {0,z},{1,z}, for z outside {0,1}; both cannot lie in P_2. If d and c are both outside 01, its unique preimage pair is disjoint from {0,1}, again impossible in P_2. Hence d is 0 or 1. If $d = 1$, examine 02. When c is in 02, its two preimages {0,z},{1,z} cannot both belong to P_3. When c is outside 02, its unique preimage pair is disjoint from {0,1} and cannot belong to P_3. Thus $d = 0$. No state therefore maps to 0 under a singular letter.

Every letter either fixes 0 with no other incoming edge, or omits 0 entirely. State 0 is unreachable from every other state, contradicting strong connectivity. The argument works for any alphabet size and any n where this type is defined; it gives only a necessary exclusion.

A.1.1 Triangle early-pair obstruction at codimension three

Claim. For $n \geq 6$, a strongly connected synchronizing automaton with an (n-3)-subset of shortest compression length ten cannot have

```
P1={01}, P2=\{01,02\}, P3=\{01,02,12\}.
```

The alphabet is arbitrary and finite; reachability of the subset is not assumed. Throughout $C = \{0, 1, 2\}$. *A* hole set covers P_3 exactly when it is C or is $\{i,j,x\}$, with $\{i,j\}$ a core pair and x outside C. Every image reached within six letters must have such holes, else a P_3 word compresses the original subset within nine letters.

The distinct images reached within i letters grow strictly at each step until a compressed image is included. *A* plateau would be invariant under every letter and contradict synchronization. Thus the assumptions imply at least i+1 distinct images within i letters for 0<=$i \leq 9$.

Letter structure The credited short-pair refinement forces every letter to be a permutation or a rank-(n-1) map with common kernel 01. Inverse pairs of 01 lie in P_2; inverse pairs of 02 lie in P_3. *A* permutation therefore preserves C. *A* singular letter has missing output 0 or 1. If it omits 1, inverse pairs of 02 force it to map C onto {0,2}. If it omits 0, write its kernel image $c = f(0)$=$f(1)$ and its other core image $t = f(2)$, with $c \neq t$ and c,$t \neq 0$.

Some permutation g satisfies g^{-1}(01)=02, since singular letters have no two-point preimages of 01. In particular $g(02) = 01$ and g preserves C and its complement. Strong connectivity requires a singular letter that does not preserve C: otherwise every letter preserves this proper set. Such a letter omits 0. Its possibilities are:

- A: c in {1,2}, t outside C;
- B_1: c outside C, $t = 1$;
- B_2: c outside C, $t = 2$;
- C: c,t outside C.

Case *A*: pure exclusion For Case A, the domain D_f of P_3-covering holes whose f-images also cover P_3 consists of C, {0,2,v}, {1,2,v}, and all {0,1,x}, where v is the unique external preimage of the other point in {1,2}. Its intersection with g^{-1}(D_f) has at most four elements. Because g sends 01 to a different core pair, at most one hole 01x survives its test, namely $x = g$^{-1}(v); the other three holes contribute at most three. All images reached within four letters must be in that intersection, contradicting strict image growth.

Case B_1: pure exclusion Let v be the unique external preimage of 2. Then

```
D_f={C,{0,1,v}}
    union {{0,2,x},{1,2,x}: x outside C}.
```

Define $D = D_f$ intersection g^{-1}(D_f) intersection f^{-1}(D_f). All images reached within four letters lie in D: appending f, gf or ff must still cover P_3 within six letters.

The core pair 02 has at most one external choice in D, because g sends 02 to 01. For pair 12, membership in f^{-1}(D_f) forces $x = v$ or $f(x) = v$, so there are at most two external choices. The pair 01 has at most its single choice v. The hole C belongs to f^{-1}(D_f) only if $c = v$. If $c = v$,

there is no external preimage of v (it is the kernel image), so the pair 12 has only its choice v. Thus $|D| \leq 4$ in both cases $c = v$ and $c \neq v$. Strict image growth within four letters gives a contradiction.

This argument excludes B_1 for every singular letter in any putative tight triangle automaton. It is subsequently used as a necessary cut.

Case B_2: reduction to two four-image domains Now $f(2) = 2$. Let $v = f\hat{}\{\text{-}1\}(1)$ outside C and let $u = g\hat{}\{\text{-}1\}(v)$, also outside C. Again D_f consists of C, $\{0,1,v\}$, and all $\{0,2,x\}$,$\{1,2,x\}$. Put

```
D=D_f intersection g^{-1}(D_f)
      intersection f^{-1}(D_f) intersection (fg)^{-1}(D_f).
```

The tests correspond to f,gf,ff,fgf, all of length at most three. Every image reached within three letters therefore lies in D.

The following direct restrictions hold:

- C is in D only if $c = u$ (test fgf);
- $\{0,1,v\}$ is in D only if $c = v$ (test ff);
- core pair 02 has at most external choice u (test gf); it further requires $u = v$ or $f(u) = u$ (test fgf);
- core pair 12 has at most external choices v and $w = f\hat{}\{\text{-}1\}(u)$, where w is included only if it exists outside C (test fgf).

If $c = u$, w does not exist outside C, since u is then the kernel image. If $c \neq u$, C is absent. These restrictions give $|D| \leq 4$. Strict growth forces equality and all four domain images to be reached within three letters. Equality has only two possibilities:

1. $c = u{=}v$. Then D is $\{012,01v,02v,12v\}$.
2. $c = v{!=}u$ and $f(u) = u$. Then D is $\{01v,02u,12v,12u\}$.

Every letter must send every image in D to a P_3-covering image: each was reached within three letters. In particular the letter assumptions A and B_1 are excluded by the pure arguments above.

For the equal-parameter B_2 domain, the occupied core rows are the four singletons of $C \cup \{v\}$, with a common J. The four rows are independent and contain R_3. If J enters a core point under f, every row of V_2D_f is proportional to the same occupied-core singleton row. But $\dim(V_2D_f) \geq 2$, a contradiction. Hence J is invariant, contradicting strong connectivity.

For the other B_2 domain, relabel $v = 3, u = 4$ and write its occupied rows $X_1 = J \cup 24$, $X_2 = J \cup 13$, $X_3 = J \cup 04$, $X_4 = J \cup 03$. These four rows are independent. The sole kernel direction of a singular letter is $k = e_0 - e_1 = X_4 - X_2$. A three-row subspace contains k exactly when its rows include X_2, X_4. If some letter f sends a point of J to a core point z, the output space V_2D_f is spanned by at most two domain rows containing z. Thus f is singular, $\dim V_2 = 3$, and R_2 consists of X_2, X_4 and one further row. At least two possible output rows must contain z. Permutations and missing-1 letters preserve 012, hence J can enter only 3 or 4 under those letters. Missing-0 letters have only one available row at 1 or 2; their rows without 0 are $24, 13$, disjoint, so every entry is impossible. For a missing-1 letter entry at 3 permits only 03. The only remaining possibility is missing-1 and $z = 4$, with outputs $24, 04$. Its kernel image is in $\{0, 2\}$, hence $X_2f = X_4f$. If the third row is X_3, it also contains 0, so all three outputs coincide. If it is X_1, then $X_1 \cup X_4$ excludes 1 and f is injective there, giving $|X_1f \cap X_4f| = |J|$. Equal output rows instead have intersection size $|J| + 2$, and distinct rows $24, 04$ have intersection size $|J| + 1$. Both alternatives contradict injectivity. Thus J has no entry into the core and is invariant, again impossible.

Case C: a six-image domain Let v_1,v_2 be the distinct external preimages of 1,2. The domain D_f is exactly the six holes formed by one core pair and one of v_1,v_2; C itself is not in D_f. Every image reached within five letters lies in D_f, because its f-image reached within six letters must cover P_3. Strict growth gives all six domain images by depth five. Every letter sends every domain image to a P_3-covering image.

The six occupied rows are the triangle-C low-rank family listed above, so the trajectory-rank contradiction excludes this case.

A.1.2 Path early-type-2 exclusion at codimension three

Claim. For $n \geq 6$, a strongly connected synchronizing automaton cannot have an (n-3)-set of shortest compression length ten and early merging pairs $P_1 = 01$, $P_2 = \{01, 02\}$, $P_3 = \{01, 02, 13\}$. The alphabet is arbitrary finite; reachability of the set is not needed for this necessary exclusion.

The short-pair refinement and its hypotheses are in Necessary short-pair structure for tight compression and Necessary early-pair types for tight compression. They force every letter to be a permutation or a rank-(n-1) transformation with common kernel 01. Every inverse pair of 01 lies in P_2 and every inverse pair of 02 lies in P_3. Write $C = \{0, 1, 2, 3\}$; an outside state means a state outside C.

Images, domains and the accounting of lengths The three-hole sets covering P_3 are exactly the four triples in C and $01x$,$03x$,$12x$ for outside x. Let H be this family. Every image of the initial set reached within six letters has holes in H; otherwise a further word of length at most three compresses the initial set within nine letters.

Let R_i be the distinct images reached within i letters. Until compression occurs this family grows strictly: equality $R_i = R_(i{+}1)$ would make it invariant under every letter, and a reset word could not compress any member. Thus $|R_i| \geq i + 1$ for $i \leq 9$.

For a finite list W of test words of maximum length L, let D(W) consist of H in H for which every test word preserves the set size and produces holes again in H. Then R_{6-L} is contained in D(W). Consequently $|D(W)|$ must be at least 7-L. Failure of a test from D(W) gives compression in at most L+3 further letters (or already compresses during the test). Hole actions below always mean complements of actual images, not preimages of subsets.

Compatible letters Every singular letter omits 0. Indeed its missing output must be 0 or 1, because otherwise inverse pairs of 01 either are disjoint from 01 or contain both $0z$ and $1z$, incompatible with P_2. If it omitted 1, inverse pairs of 02 would similarly either be disjoint from 01 or contain both $0z$ and $1z$. Neither possibility belongs to P_3. Write such a letter f as $f0 = f1$=c, $f2 = t$, $f3 = z$, where c,t,z are distinct and nonzero.

Permutations fall into three classes, by their specified core rows:

- A: (0,1,2,*) on C;
- B: (0,2,1,*) on C;
- T: (,,,) with $T_0 = 1$, $T_1 = 0$, $T_3 = 2$.

The free image in B is 3 or outside C; that in T is $T_2 = 3$ or outside C. Call these closed and mixing respectively. There is a B permutation: the new inverse pair 02 of 01 must arise from a permutation, and class T has inverse pair 01 instead. There is also a T permutation: only it has inverse pair 13 of 02. Singular letters omit 0, so create neither of those new inverse pairs. The

classification follows directly by intersecting the inverse pairs of 01 and 02; their common inverse endpoint is the preimage of 0.

Permutation mixing Direct hole substitution gives the following domain bounds. Juxtaposition of letters is right action, so TB means first T, then B.

Letters	Tests W	L	Maximum	Core holes and outside restrictions
B mixing, T mixing	B,T	1	5	012,023; at most one outside choice for each of 01,03,12
B mixing, T closed	B,TB	2	5	023,123; $01v$,$03v$,$12u$, where $v = B^{-1}(3)$, $u = T^{-1}(v)$
B closed, T mixing	B,T,TT	2	4	At most three core holes and one outside hole, or two core holes and two outside holes
B,T closed, A mixing	A,B,TA	2	4	012,013; at most $03v$ and $12T^{-1}(v)$, $v = A^{-1}(3)$

In the first row $v = B^{-1}(3)$ fixes the outside choice for 01 and 03, and $T^{-1}(3)$ fixes it for 12. The bound five contradicts $|R_5| \geq 6$.

In the second row, membership of $01v$ further requires $T(v) = v$. Equality five therefore forces $u = v$, and its domain is exactly

`D={023,123,01v,03v,12v},  v outside C, T(v)=v.`

If its size is smaller than five, $|R_4| \geq 5$ gives a contradiction. The equality case is treated by the five-cycle argument below.

For the third row put $a = T_2$ outside C and $v = T^{-1}(3)$. If $v = a$, the possible core holes are 012,013,023 and the only outside hole is $12v$: $T^{-2}(3) = 2$ is in C. If $v \neq a$, only 012,013 remain among core holes, and at most $12v$,$03T^{-2}(3)$ remain outside. Both give the stated four. In the last row A preserves 012 and sends 3 outside. The B test deletes all $01x$; the A and TA tests restrict the other two types to the listed unique preimages. These two rows contradict $|R_4| \geq 5$.

The equality domain has the type-2 five-cycle occupied rows listed above, so the common-entry argument excludes it. Thus all permutations preserve C.

Singular mixing once permutations preserve C Choose a closed B and T. Strong connectivity requires a singular f that does not preserve C. Retain $f0 = f1$=c, $f2 = t$, $f3 = z$. Say t or z is good when it belongs to {1,3}. Direct substitution gives:

Good images among t,z	Tests	L	Domain bound
neither	B,f	1	4
exactly one	B,f,Tf	2	4
both	B,fB,TfB	3	4

For the first row B excludes $01x$. The holes 023,123 fail f. If c is 1 or 3, the two other core holes may remain, but only the other member of {1,3} has an outside preimage; each of $03x$ and $12x$ has at most that one choice. If c is neither, no core hole remains and each outside type has at most two choices. Both counts are at most four.

In the second row T swaps 012 with 013, so the tests f,Tf retain either both of these holes or neither. The holes 023,123 are similarly paired. Each outside type has at most the unique outside preimage of the other member of {1,3}. If all four core holes remain, c must be that other member, so this outside preimage does not exist. Thus again the count is at most four. The first two rows contradict respectively $|R_5| \geq 6$ and $|R_4| \geq 5$.

In the last row $\{t,z\}=\{1,3\}$; mixing forces c outside C. Let $v = f^{-1}(2)$, $u = T^{-1}(v)$, both outside C. The domain has exactly FOUR holes:

```
t=1,z=3: D={023,123,12v,03u};
t=3,z=1: D={023,123,03v,12u}.
```

It is not a two-image domain. Both 023 and 123 have f-image holes 013, which is outside D, so two distinct vertices have exit distance one. Every vertex must exit under synchronization. In a four-vertex graph with two distinct distance-one vertices, maximum exit distance is at most three: a shortest distance-four path would use distinct vertices of all four levels, leaving only one vertex at level one. The initial vertex is in D. Its exit takes at most three letters; a failed defining test takes at most three more, and a P_3 merge at most three more. Thus compression has length at most nine. This excludes the last singular mixing possibility and contradicts strong connectivity.

A.1.3 Path early-type-3 exclusion at codimension three

Claim. For $n \geq 6$, a strongly connected synchronizing automaton cannot have an (n-3)-set of shortest compression length ten and

```
P1={01}, P2=\{01,02\}, P3=\{01,02,23\}.
```

The alphabet is arbitrary finite, and reachability of the set is not needed. Necessary short-pair structure for tight compression supplies the credited short-pair reduction. Let $C = \{0, 1, 2, 3\}$. Three-hole sets covering P_3 form the family H consisting of the four triples in C and $02x$,$03x$,$12x$ for x outside C.

As in Path early-type-2 exclusion at codimension three, every image reached within six letters has holes in H. Distinct reachable images grow strictly until compression, so $|R_i| \geq i + 1$ through depth nine. For tests of maximum length L, their common H-preserving domain must contain R_{6-L}, hence have at least 7-L vertices. Exiting that domain gives compression after at most L+3 further letters. These length charges always use original letters.

Compatible letters Every letter is a permutation or a rank-(n-1) map with common kernel 01. Its inverse pairs of 01 belong to P_2, and those of 02 belong to P_3. The permutation templates are:

```
A: (0,1,2,*) on C;
B: (0,2,1,*) on C;
T: (1,*,0,2) on C.
```

A,B are closed when their last core image is 3; T is closed when $T_1 = 3$. Otherwise that image is outside C. Singular letters either omit 0, with $f0 = f1$=c, $f2 = t$, $f3 = z$ distinct and nonzero, or omit 1 with one of

```
D0: (c,c,0,2),  D2: (c,c,2,0),  c>=3.
```

These last letters are closed when $c = 3$, otherwise mixing. To check completeness, a singular missing output is 0 or 1 by the inverse-pair 01 test. If it omits 1, its kernel image cannot belong to 02: that would create both $0x$ and $1x$ as inverse pairs, which cannot both lie in P_3. The inverse pair of 02 must therefore be the disjoint pair 23, giving the two displayed orientations. The permutation templates follow by intersecting the inverse pairs of 01 and 02. Growth from P_1 to P_2 requires B or T; growth from P_2 to P_3 requires T or D_0 or D_2.

A mixing T permutation Put $a = T_1$ outside C and $v = T^{-1}(3)$. Test T and TT. If $v \neq a$, their common domain is contained in $\{02v,12v,03v,03T^{-1}(v)\}$, of size at most four, contradicting $|R_4| \geq 5$. If $v = a$, the domain is exactly

```
D={013,123,02v,12v,03v}.
```

The occupied core rows are the type-3 five-cycle family listed above. The common-entry argument excludes mixing T.

A closed T permutation Now T on C is (1,3,0,2). Its outside action is a permutation. For an additional letter g, let $D_2(T, g)$ be the domain preserving H under all words of length at most two on T,g. Direct substitution gives:

Mixing g	Domain is contained in	Definition of v,u
A	123,03v,12u	$v = g^{-1}(3)$, $u = T^{-1}(v)$
B	012,03v,12u	same
D_0	013,03u,12v	same
D_2	023,123,12u	same

For A, the tests g,gT,Tg remove respectively core 013,023,012; for B they remove 023,013,123. Their outside constraints require $gx = 3$ for $03x$ and $g(Tx) = 3$ for $12x$. In the D_0 case gg removes 023,123 and g removes 012; its 013 survives only if $g(g0) = 3$, equivalently $g0 = v$, while the outside restrictions give the displayed unique preimages. In D_2, g removes 013, Tg removes 012, and gg removes every possible $03x$. These observations verify all four rows. Every domain has size at most three, contradicting $|R_4| \geq 5$. Thus every permutation and every missing-1 letter preserves C. Strong connectivity requires a mixing missing-0 letter f with core images c,t,z.

Call a core output good if it is 2 or 3. If at most one of c,t,z is good, $D_2(T, f)$ has at most four vertices. With none good there are no core holes, and each outside type $03x$,$12x$ has at most the

two outside preimages of 2,3. With only c good, there is at most core 012 and one outside choice for each type. With only t good, core holes are among 023,123, with at most $03v$,$12T^{-1}(v)$; with only z good they are among 013,123, with at most $12v$,$03T^{-1}(v)$. Here v is the outside preimage of the other good output. These counts are at most four.

If c and one of t,z are the two good outputs, the remaining core image is outside C (otherwise f is not mixing). All outside holes are removed by f and Tf. At least one core hole is removed by fT: if $c = 2$ it is 013 when z is outside, or 012 when t is outside; if $c = 3$ the holes 023,123 are removed. The domain then has size at most three.

It remains that c is outside and $\{t,z\}=\{2,3\}$. Set $v = f^{-1}(1)$ and $u = T^{-1}(v)$, both outside C. Compute $D_3(T, f)$, the domain preserving H under all words of length at most three:

```
t=2,z=3: D3={013,123,03u};
t=3,z=2: D3={023,123,03v}, plus 12u iff f(u)=v.
```

For completeness the two-step domains before the last tests are

```
t=2,z=3: {013,023,123,12v} union {03x: x outside C};
t=3,z=2: {012,023,123,03v} union {12x: x outside C}.
```

Requiring their T- and f-successors to remain in those domains gives the displayed D_3 sets directly. Their sizes must be at least four because $|R_3| \geq 4$. Only $t = 3$,$z = 2$ and $f(u) = v$ survive. Since $f(v) = 1$, $u \neq v$. Relabel $u = 4$,$v = 5$ and order the remaining domain as

```
D=\{023,123,124,035\}.
```

Here $T = (1, 3, 0, 2, 5, \ldots)$ and $f = (c, c, 3, 2, 5, 1, \ldots)$, with distinct unspecified outputs and $c \notin \{0, 1, 2, 3, 5\}$. For the holes $023, 123, 124, 035$, respectively, apply T, TT, ff, f. Each yields holes 012, outside the domain, without intermediate compression. For example $124f = 035$ and $123T = 023$. Thus every domain row exits within two letters; a failed defining test costs at most three and a P_3 merge at most three more, giving compression in at most eight letters.

No T permutation There must now be a B permutation and a D_0 or D_2 letter. A closed D_0 is impossible even alone: its common domain for D_0,D_0D_0 is precisely the four core holes, fewer than $|R_4| \geq 5$. A mixing D_0 together with B also has a two-step domain of size at most four. To see this write $v = D_0^{-1}(3)$. The D_0 and D_0D_0 tests leave at most core 013, only if $D_0(v) = 3$ with $v = D_0(0)$, and outside holes $12v$,$02v$,$03x$ with $x = v$ or $x = D_0^{-1}(v)$. If B is closed it removes $02v$; if core 013 is present, $D_0^{-1}(v)$ is not outside, giving at most three vertices. If B is mixing, its test forces $x = B^{-1}(3)$ for $03x$ and for $02x$. There are then at most one core hole and three outside holes. Both counts are at most four. Thus no D_0 letter exists, and a D_2 letter is necessary.

If the alphabet consisted solely of A,B permutations and D_2 letters, the family of holes $12x$ ($x \neq 1,2$, including $x = 0,3$) would be invariant. A,B preserve the pair 12. A D_2 letter sends 120 to 12c and any other $12x$ to $12D_2(x)$; these are again three distinct holes. This invariant family of uncompressed (n-3)-images contradicts synchronization. There is therefore a missing-0 letter f.

Choose one $B = b$, one $D_2 = d$ and such an f. Let beta be 1 when b is mixing, epsilon be 1 when d is mixing. In their common H-domain, $02x$ has at most the single outside choice $b^{-1}(3)$, available only when $beta = 1$; $03x$ has at most the outside choice $d^{-1}(3)$, available only when $epsilon = 1$. Core 023 is absent if $beta = 1$, and core 013 is absent if $epsilon = 1$. Let $c = f0$=f1, $t = f2$, $z = f3$.

First suppose t is neither 2 nor 3. If neither c nor z is good, all core holes fail and each outside type $02x$,$12x$ requires fx in {2,3}; there are at most two $12x$ and at most one of each other type,

totaling four. If only c is good, core holes are among 012,013, and all outside types require the unique outside preimage v of the other good output. There are at most three outside holes; if all three are present, both b and d are mixing and core 013 is absent, again totaling at most four. If only z is good, core holes are among 013,023,123; the possible $02x$,$03x$ are paid for by the respective losses of 023,013, leaving at most three core/outside-extra holes plus the one possible $12x$. If both c,z are good, $02x$ and $12x$ are impossible since neither good output has an outside preimage. The possible $03x$ is paid for by the absence of core 013. In each case the one-step domain has size at most four.

Now suppose $t = 2$ or $t = 3$ and use all tests of length at most two on b,d,f. Define a two-element output-target set V by

```
t=2: V={1,b^-1(3)};
t=3: V={2,w}, w=1 if d is closed, w=d^-1(3) if d is mixing.
```

The test fb in the first case, or fd in the second, forces fx in V for every allowed outside $12x$. Put a=[c in V], h=[z in V]. The kernel output c and core output z have no outside preimages, so at most 2-a-h outside $12x$ survive. The same test requires c in V for core 012, and z in V for core 023,123. Hence the core count is at most

```
a + h(2-beta) + (1-epsilon).
```

The last term bounds core 013, and the beta term removes 023 under mixing B. The other two outside types contribute at most beta+epsilon. The total domain size is therefore at most

```
a+h(2-beta)+(1-epsilon)+(2-a-h)+beta+epsilon
   = 3+h(1-beta)+beta <= 4.
```

This also covers the common-kernel issue in fd: when d is closed, preimages of output 3 are 0,1, but f omits 0, leaving only target 1. When d is mixing the preimage is the unique outside state $d^{-1}(3)$. All targets in V are distinct and neither equals t.

Thus the common two-step domain on b,d,f has at most four images, contradicting $|R_4| \geq 5$. This completes the no-T branch and the claim.

A.1.4 Disjoint-tail early-type-4 exclusion at codimension three

Claim. For $n \geq 5$, a synchronizing automaton cannot have an (n-3)-set of shortest compression length ten with

```
P1={01}, P2=\{01,02\}, P3=\{01,02,34\}.
```

No alphabet-size, strong-connectivity or image-reachability hypothesis is needed for this exclusion. Necessary short-pair structure for tight compression gives the credited short-pair refinement. Write $C = \{0, 1, 2, 3, 4\}$ and $G = \{3, 4\}$. The three-hole sets covering P_3 form H: the seven core holes

```
013,014,023,024,034,123,124
```

and $03x$,$04x$ for x outside C. Every image reached within six letters must have holes in H. The distinct images reached within i letters grow strictly until compression, giving $|R_i| \geq i + 1$ for $i \leq 9$. Equality at a step would make all reached uncompressed images invariant under every letter, contrary to the existence of a reset word.

All compatible letters Every letter is a permutation or a rank-(n-1) transformation with common kernel 01. Inverse pairs of 01 belong to P_2 and inverse pairs of 02 belong to P_3. Permutations have core prefix A=(0,1,2) or B=(0,2,1); they fix 0 and preserve the pair 12. An inverse pair 34 of 02 cannot occur in a permutation: its inverse pair of 01 is 01 or 02 and would be disjoint from 34, whereas the target pairs intersect at 0.

A singular letter omits 0 or 1, by the inverse-pair 01 test. If it omits 1, its kernel image cannot be 0 or 2: that would produce both $0x$ and $1x$ as inverse pairs of 02, incompatible with P_3. Its unique inverse pair of 02 must therefore be 34. It has one of the core rows

```
(c,c,t,0,2), (c,c,t,2,0),  c,t>=3, c!=t.
```

A singular letter omitting 0 has unrestricted distinct nonzero images on the common kernel and the other core states. Some missing-1 letter is necessary, because only such a letter can add the new pair 34 to P_3. Call it d. Swapping 3 and 4 if needed, write d=(c,c,t,0,2) on C. This relabeling preserves the whole early-pair type.

The domain of d has six images Let D be the holes in H whose d-images also have holes in H. The hole 034 always has d-image holes 012, so is absent. Core holes 013,014 survive exactly when c is in G; holes 023,024,123,124 survive exactly when t is in G. An outside $03x$ or $04x$ survives exactly when dx is in G. Among the two outputs in G, every one not occupied by c or t has one outside preimage. Thus, writing a=[c in G], b=[t in G],

```
|D|=2a+4b+2(2-a-b)=4+2b.
```

This exact count also follows by direct complement-of-image substitution. For example, 013 has holes {1,c,0}, 014 has {1,c,2}; 023 and 123 have {1,t,0}, 024 and 124 have {1,t,2}; outside holes have {1,0,dx} or {1,2,dx}. In each case the remaining condition is to meet G.

All images reached within five letters belong to D; hence $|D| \geq 6$. We must have t in G and $|D| = 6$, so every D-image is reached within five letters. In particular EVERY letter sends EVERY D-image to an H-image, because the resulting depth is at most six.

There are exactly two domain shapes:

- c in G: c,t are the two members of G and $D_closed = \{013, 014, 023, 024, 123, 124\}$;
- c outside C: let v be the unique outside preimage under d of the other member of G. Then $D_mixing = 023, 024, 123, 124, 03v, 04v$.

The second shape requires $n \geq 6$. The first is already meaningful at $n = 5$.

Both occupied row families are the type-4 low-rank families listed above; their rank is at most four, contradicting the trajectory rank.

A.1.5 Crossing early-type-5 exclusion

This is one elementary case of Uniform codimension-three compression bound.

Claim. No automaton can have exactly

```
P1={01}, P2=\{01,23\}, P3=\{01,23,02\},
```

where P_h denotes the unordered distinct-state pairs merging in at most h letters. No strong-connectivity, synchronization, alphabet-size or reachable-image assumption is needed. The labels require $n \geq 4$.

The single pair in P_1 forces each letter to be a permutation or a rank-(n-1) map with common kernel 01. Pair 02 is new at level three, so some letter f must send it to a distinct pair in P_2. It cannot send it to 01, since that would put 02 in P_2. Thus $02f = 23$ is necessary.

If f is a permutation, its inverse pair of 01 is either 01 or 23 by the definition of P_2. Its inverse pair of 23 would have to be 02. Inverse pairs of the disjoint target pairs 01 and 23 must be disjoint, but 02 intersects BOTH possible inverse pairs of 01. This is impossible.

If f is singular, its common kernel is 01. For $02f = 23$, f0 is an endpoint of 23 and f2 the other endpoint. Since $f1 = f0$, also $12f = 23$. This would put the additional pair 12 in P_3, contradicting the three specified pairs. The two cases exhaust every letter and prove the claim.

A.1.6 Kernel-tail early-type-6 exclusion at codimension three

Claim. For $n \geq 6$, a strongly connected synchronizing automaton cannot have an (n-3)-set of shortest compression length ten with

`P1={01}, P2=\{01,23\}, P3=\{01,23,04\}.`

The alphabet is arbitrary finite and the set need not be reachable. The credited short-pair refinement is in Necessary short-pair structure for tight compression. Let $C = \{0, 1, 2, 3, 4\}$. The three-hole P_3 covers form H: core holes 012,013,023,024,034,124,134 and outside holes $02x$,$03x$, x outside C. Every image reached within six letters has holes in H. The distinct images grow strictly until compression (a plateau would be invariant under all letters and contradict synchronization), hence $|R_i| \geq i + 1$ for $i \leq 9$. Tests of maximum length L have a common H-preserving domain containing R_{6-L}. If a word leaves that domain, a failed test and a P_3 word compress in at most L+3 additional letters.

Complete letter classification Each letter is a permutation or a rank-(n-1) map with common kernel 01. Its inverse pairs of 01 belong to {01,23}, and inverse pairs of 23 belong to {01,23,04}. Permutations have precisely three forms:

- A: preserve each pair 01,23;
- B: exchange those two pairs;
- T: map 23 onto 01 and 04 onto 23, with the image of state 1 outside {0,1,2,3}.

The first two have core images in {0,1,2,3} and arbitrary outside images. In T the four images of 0,2,3,4 are all of {0,1,2,3}.

A singular letter cannot have its kernel image in a target pair whose inverse pairs are required: that would give both $0x$ and $1x$, which cannot both lie in either allowed early-pair family. If its missing output is outside 01, it must map 23 onto 01. Its missing output must then be 2 or 3, since otherwise its inverse pair of 23 could be none of the permitted pairs. If it omits 0 or 1, it must map 23 onto 23. Thus all singular letters have one of these forms:

- U: missing e in {0,1}, map 23 onto 23, kernel image c in {1-e} union {4,5,...};
- V: missing e in {2,3}, map 23 onto 01, kernel image c in {5-e} union {4,5,...}.

Their images of other states are distinct and omit e and c. Only T can create the new pair 04, so some T permutation is necessary. Swapping labels 2 and 3, the chosen T has one of the normalized core rows

```
I:  (2,a,0,1,3),   II: (2,a,1,0,3),   a>=4.
```

Call it closed when $a = 4$ and mixing otherwise. The swap fixes 0,1,4 and preserves the ordered early-pair type.

Orientation II If $a \geq 5$, put $v = T^{-1}(4)$. The common domain for T and TT is exactly

```
a=v:  \{013,034,134\};
a!=v: {034,134,03w},  w=T^-1(v) outside C.
```

For example the one-step domain is 013,023,034,134,02v and all outside 03x. The second test removes 023 and 02v; it retains 013 only if $a = v$, and an outside 03x only if $Tx = v$. Since $T_1 = a$, exactly one of these last possibilities exists. The domain has size three, fewer than $|R_4| \geq 5$, so mixing orientation II is impossible.

If $a = 4$, the T,TT domain is precisely the five core holes

```
D=\{012,023,034,124,134\}.
```

Its occupied rows form the type-6 five-cycle listed above. The common-entry argument excludes closed orientation II.

Orientation I requires a breaker Let T=(2,a,0,1,3) on C, put $v = T^{-1}(4)$, and sigma=[$a = v$]. The domain for T,TT is exactly

```
closed: \{012,013,023,024,134\} union {02x: x outside C};
mixing: {012,023,024,03v} union {02x: x outside C},
        with 124 added only when sigma=1.
```

In the closed case $v = 1$. In the mixing case v is outside C. These identities follow by applying T to each of the seven core holes and to 02x,03x separately; the one-step mixing domain is {012,023,024,124,03v} union {02x: x outside C}.

For a permutation g, call it a keeper when {g0,g2}={0,2}. For a singular g with missing output e, call it a keeper when {e,g2}={0,2}. If every letter were a keeper, the family of holes 02x for all $x \neq 0,2$ (including $x = 1,3,4$) would be invariant. For singular letters, the hole 012 maps to 02c, and every other 02x maps to 02gx; each is again three distinct holes. Thus these (n-3)-images could never compress, contradicting synchronization. There is a breaker g.

All orientation-II letters have already been excluded, so the complete breaker list is as follows. Below h denotes g4, except in the T row where $h = g1$; each free output obeys the rank and missing-output rules.

Name	Core row or conditions
A_{03}	(0,1,3,2,h)
B_{03}	(3,2,0,1,h)
A_bad	(1,0,k,l,h), {k,l}={2,3}

Name	Core row or conditions
B_bad	$(k,l,1,0,h)$, $\{k,l\}=\{2,3\}$
T_other	$(3,h,1,0,2)$, $h \geq 4$
U_{03}	(c,c,3,2,h), missing 0
V_{03}	(c,c,0,1,h), missing 3
U_bad	U missing 1
V_bad	V with $g2 = 1$, $g3 = 0$

This is exhaustive from the A,B,T,U,V classification by checking whether the carrier of the holes $02x$ is 02,03,12 or 13. Carrier 02 is precisely the keeper case. It is important that carrier 12 or 13 also covers P_3 when its third point is 0, as well as when it is 4; that boundary is retained in the counts below.

Uniform two-step counting ledger Let D(T,g) be the H-domain of ALL words of length at most two on T,g. Its core holes are among the displayed T,TT core holes. Its outside holes are only $02x$ and, in the mixing case, at most the single $03v$. Write C_i=$[c = i]$, H_i=$[h = i]$. Each triple in the tables bounds

`(number of core holes, number of outside 02x, number of outside 03x).`

The closed-I ledger is:

Breaker	Component bounds
A_{03}	(4,0,0)
B_{03}	(3,0,0)
A_bad, B_bad	(3+H_4,1-H_4,0)
T_other	(2+H_4,1-H_4,0)
U_{03}	(3+C_1+H_1,1-C_1-H_1,0)
V_{03}	(1+C_2+H_2+H_4,1-C_2-H_2,0)
U_bad	(2C+2H,2-C-H,0), $C = C_0$+C_4, $H = H_0$+H_4
V_bad	(3+C_4+H_4,1-C_4-H_4,0)

Every sum is at most four. For U_bad, c,h are distinct so C,H are each indicators, giving sum $2+C+H \leq 4$. Negative-looking terms are nonnegative under the distinct-output conditions.

The mixing-I ledger is:

Breaker	Component bounds
A_{03}	(1+Hv+sigma H_4,1-Hv,H_4)
B_{03}	(1+Hv+sigma,1-Hv,0)
A_bad	(1+H_4+sigma(1-H_4),1-H_4,1-H_4)
B_bad	(1+H_4(1+sigma),1-H_4,H_4)
T_other	(sigma H_4,1-H_4,H_4)
U_{03}	(1+Cv+Hv(1+sigma),1-Cv-Hv,1)
V_{03}	(C+H(1+sigma),2-C-H,0), $C = C_2$+Cv, $H = H_2$+Hv

Breaker	Component bounds
U_bad	(C+H(1+sigma),2-C-H,1), $C = C_0$+C_4, $H = H_0$+H_4
V_bad	(1+C_4+H_4(1+sigma),1-C_4-H_4,H_4)

Here $v \geq 5$, so the indicator sums in the V_{03} row are again indicators. Every row sums to at most four (some have the smaller upper three).

To verify the ledger algebraically, apply the hole formula given below to the core holes and the two outside types. The restrictions on outside $02x$ reduce to unique preimages of these target sets:

- permutation bad rows: {4};
- U_{03}: $\{v\}$, with $v = 1$ in the closed case;
- V_{03}: {2,v}, with target 1 already having core preimage 3 when closed;
- U_bad: {0,4};
- V_bad: {4}, since output 0 already has core preimage 3.

Every target occupied by kernel image c or core image h has no outside preimage. This gives all the negative terms in the second column. For permutation A_{03}/B_{03} the targets of gT on $02x$ are {2,v}, but output 2 already has a core preimage, leaving only v; when closed, output $v = 1$ also has a core preimage, leaving none.

Core counts are obtained by substituting the five/four candidate core holes of T,TT into g,gT,Tg,gg. For illustration, U_{03} has three possible unconditional closed core holes 013,023,134 and the two extra holes 012,024 only if $c = 1$,$h = 1$. In the mixing case it has unconditional 023, 012 only if $c = v$, and 024 (also 124 if $sigma = 1$) only if $h = v$.

For V_{03} the closed hole 013 cannot survive: its first image would require $c = 4$ and be 134, but its second image then requires $h = 4$, contradicting distinct outputs. The remaining core candidates are 023,012 if $c = 2$, 024 if $h = 2$ and 134 if $h = 4$. In the mixing case 023 fails gT, and $03v$ also fails gT: its possible first images 013 or 134 both fail T. The displayed C,H restrictions on 012,024,124 then give its table row. The bad U core counts simply test whether c,h belong to {0,4}; the bad V rows have three unconditional possible closed core holes, with 012/024 requiring $c = 4$,$h = 4$. In the mixing bad V case only 023 is unconditional, 012 requires $c = 4$, and 024, the optional 124 and $03v$ require $h = 4$. The permutation rows follow from their explicit core templates in exactly the same substitution.

The hole formula is: a permutation sends holes J to $g(J)$; a singular common-kernel letter missing e sends J to $\{e\}$ union $g(J)$ if J contains both 0,1, and to $\{e\}$ union $g(J$ minus $\{0,1\})$ otherwise. In particular it correctly includes the kernel image when BOTH kernel points are holes. This proves the entire finite ledger for arbitrary n, without extrapolation from numerical instances.

It follows that $|D(T, g)| \leq 4$. But every image reached within four letters belongs to it, whereas $|R_4| \geq 5$. This contradiction excludes orientation I and completes the claim.

A.1.7 Second-pair-tail early-type-7 exclusion at codimension three

Claim. For $n \geq 6$, no strongly connected synchronizing automaton has an (n-3)-set whose shortest compression length is ten and whose early pairs are

P1={01}, P2=\{01,23\}, P3=\{01,23,24\}.

The alphabet is arbitrary finite; reachability of the set is unnecessary. This claim alone is one case of the complete nine-type exclusion, rather than an asserted all-parameter avoid-or-compress formula.

Put $C = \{0, 1, 2, 3, 4\}$. The three-hole covers H of P_3 are the seven core holes 012, 023, 024, 123, 124, 034, 134 and outside holes $02x$,$12x$, x outside C. Every image reached in at most six letters has holes in H: otherwise a pair in P_3 is still in the image and a word of length at most three compresses it before ten. The reached images R_i grow strictly until compression, since a plateau is invariant under every letter and contradicts synchronization. Thus $|R_i| \geq i + 1$ for $i \leq 9$.

For tests of maximum length L, their common domain of holes that stay in H contains R_{6-L}, and consequently has at least 7-L members. Upon leaving this domain, one failed test followed by a merging word of length at most three gives compression within L+3 additional letters.

Complete letter classification Every letter is a permutation or a rank-(n-1) map with common kernel 01. Inverse pairs of 01 must be 01 or 23; inverse pairs of 23 must be 01,23, or 24. The compatible permutation core rows are precisely

```
A: (k0,k1,l0,l1,h),
B: (l0,l1,k0,k1,h),
T: (k0,k1,l0,h,l1),
```

where (k0,k1) orders 0,1, (l0,l1) orders 2,3, and $h \geq 4$. Their outside outputs are distinct and avoid the four specified outputs.

For a singular letter, write e for its omitted output and $c = f0$=f1. If e is outside 01, the inverse pair of 01 must be 23. Its kernel image cannot lie in 01, since then both $0z$ and $1z$ would be inverse pairs. If e were also outside 23, its inverse pair of 23 would be disjoint from 23, hence could only be 01. That would put c in 23 and give two forbidden inverse pairs there. Therefore e is in 23. If e is in 01, the inverse pair of 23 is 23 or 24, with c outside 23. This gives precisely

```
U_A: missing e=0 or 1, (c,c,l0,l1,h),
U_T: missing e=0 or 1, (c,c,l0,h,l1),
V:   missing e=2 or 3, (c,c,k0,k1,h).
```

For U, c is 1-e or at least 4; for V, c is 5-e or at least 4. The other images are distinct and avoid e,c and the specified pair. These conditions also suffice for compatibility. Identity letters impose no additional restrictions; omitting them from full-row pools does not affect any domain or impossibility statement.

The seven forced domains The following are hole sets, not occupied core sets. An outside point v or w is relabeled 5.

name	hole domain	contains
$D_6closed$	023,123,024,124,034,134	R_5
D_6Tmix	024,124,034,134,02v,12v	R_5
D_6Amix	023,123,034,134,02v,12v	R_5
C_A	012,024,124,034,134	R_4
C_T	012,023,124,034,134	R_4

name	hole domain	contains
C_V	023,024,124,034,134	R_4
D_w	024,124,034,134,02w	R_4

Taking complements in the named support gives the occupied families above. All except C_T are excluded by rank at most four; C_T is excluded by the five-cycle common-entry argument. The following substitutions derive the domains exhaustively.

Eliminate all letters taking 2 to 3 For a permutation f with $f2 = 3$, its free core output is f4 in form A, or f3 in form T. If that output is 4, the one-step H-preserving domain is $D_6closed$. Otherwise $v = f^{-1}(4)$ is outside C, and the domain is D_6Amix or D_6Tmix respectively. These are checked by mapping the seven core holes and the families 02x,12x directly.

For a singular U row with $f2 = 3$ the domain size is 6-[$c = 4$]. To see the count, let z be its free core output. Hole 012 survives exactly when $c = 4$; the two holes 023,123 (form A) or 024,124 (form T) survive exactly when $z = 4$. The four other core holes survive. An outside preimage of 4 supplies two outside holes unless 4 is already c or z. Since c and z are distinct, the count is

```
[c=4]+2[z=4]+4+2(1-[c=4]-[z=4]) = 6-[c=4].
```

If $c = 4$, size five contradicts $|R_5| \geq 6$. Otherwise it is precisely $D_6closed$ when $z = 4$, or D_6Amix/D_6Tmix with $v = f^{-1}(4)$ outside C. The six-member low-rank families therefore exclude every letter with $f2 = 3$.

Eliminate V rows omitting 3 For V missing 3 put A=[c in {2,4}], B=[h in {2,4}]. Its one-step domain has size 4-A+2B: hole 012 contributes A, holes 023,123 fail, the other four core holes contribute 4B, and the outside holes contribute 2(2-A-B). If $B = 0$ it has at most four members. If $B = 1$ and $A = 1$ it has five. The only remaining case has six members, $c \geq 5$ and h in {2,4}, and is D_6Tmix with v the outside preimage of the other member of {2,4}. It too is excluded.

Eliminate B four-cycles After the preceding cuts a B permutation has $B^2(2) = 2$ or 3. In the latter case a simultaneous exchange of labels 0,1 normalizes it to

```
b=(2,3,1,0,a), a>=4.
```

For $a = 4$ the b,bb domain is C_V, and its low-rank family excludes this case. For $a \geq 5$ put $v = b^{-1}(4)$. The domain is

```
a=v:  \{023,034,024\};
a!=v: {023,034,02w}, w=b^-1(v) outside C.
```

It has only three members, contradicting $|R_4| \geq 5$.

***A* B two-cycle and a breaker are necessary** If no B permutation remains, every letter preserves the family I of three-hole sets containing 2 and at least one of 0,1. The remaining A,T permutations fix 2 and preserve 01; U maps the two carriers to {e,2}; V omitting 2 maps them to {2,f2}. The both-kernel-endpoints case also preserves I. Images with these holes never compress, contrary to synchronization. Choose a B two-cycle and normalize it by exchanging 0,1 to

```
b=(2,3,0,1,a), a>=4.
```

Call a permutation a keeper if {g0,g2}={0,2}, and a singular row a keeper if {e,g2}={0,2}. Every keeper preserves the family F of all holes 02x, $x \neq 0,2$. Thus synchronization requires a breaker. After the preceding cuts the exhaustive breaker forms are

```
P_A: (1,0,2,3,h),       P_T: (1,0,2,h,3),
P_B: (3,2,1,0,h),
U_A: missing 1, (c,c,2,3,h),
U_T: missing 1, (c,c,2,h,3),
V:   missing 2, (c,c,1,0,h).
```

Here $h \geq 4$ for permutations; c is 0 or ≥ 4 for U and 3 or ≥ 4 for V. In singular rows h avoids the missing output, c, and the specified pair.

Mixing B two-cycles If $a \geq 5$ put $v = b^{-1}(4)$. Directly testing b and bb gives the domain

```
F union {12v} union ({034} if a=v, otherwise empty).
```

In particular it has at most two members outside F. For a breaker g, write Nc for the count of core holes in D(b,g) intersect F, and No for the count of its outside holes. The following upper bounds follow by mapping the carriers of 02x under g and then under b. For U set C=[c in {0,v}], H=[h in {0,v}]; for V set C=[$c = v$], H=[$h = v$]; for P_A,P_T set H=[$h = v$].

breaker	Nc	No
P_A,P_T	1+H	1-H
P_B	0	1-[$h = 4$]
U_A,U_T	C+H	2-C-H
V	1+C+H	1-C-H

For example U converts the carriers to 12; after b these are 03. The third output must be 0 or v so that its b-image is 2 or 4. The outputs already occupied by c or h remove precisely C+H available outside preimages. The analogous V carrier is 12 with output 0 already on input 3, leaving only the target v. The bounds sum to at most two; adding the at most two non-F members gives $|D(b, g)| \leq 4$, contradicting $|R_4| \geq 5$. Hence $a = 4$.

Closed B two-cycles For $a = 4$ the b,bb domain is F union {124,034,134}. Count core and outside 02x members of D(b,g) as Nc,No. For U put C=[c in {0,4}], H=[h in {0,4}], H_0=[$h = 0$], H_4=[$h = 4$]; for V put C_4=[$c = 4$], H_4=[$h = 4$].

breaker	Nc	No
P_A,P_T	1+4H_4	1-H_4
P_B	1+3H_4	1-H_4
U_A	C+2H_0+4H_4	2-C-H
U_T	C+H_0+3H_4	2-C-H
V	3+C_4+2H_4	1-C_4-H_4

These bounds are obtained by testing the six core holes individually and counting outside preimages of the same carrier targets. Their sums are at most five; P_B and U_T have sums at most four. For U the cases $h = 0$ and $h = 4$ must be distinguished: $h = 0$ does not preserve 034,134.

Since $|R_4| \geq 5$, equality is necessary. Testing its core holes gives:

- P_A has $h = 4$ and domain C_A;
- P_T has $h = 4$ and domain C_T;
- U_A has $h = 4$. If $c = 0$ its domain is C_A; otherwise $c \geq 5$ and its domain is D_w, where $w = g^{-1}(0)$ is outside C;
- V has $h = 4$ and domain C_V.

All these five-member domains are excluded by the rank and common-entry arguments above. This exhausts every alphabet and proves the claim.

Type 8: three disjoint pairs

Here every safe image selects one vertex from each of $01, 23, 45$ and contains every state outside those six. The three block-coordinate sums are equal, and each outside coordinate equals their common sum. The row space has dimension at most four and contains no reset row $|S|e_q$ (also when $n = 6$ and the outside set is empty). The reset-row version of the plateau argument thus gives $\dim V_4 \geq 5$, whereas R_4 lies in this space. This is a contradiction. The nine cases exhaust the possible early pair layers, proving Lemma~9.

A.2 Graph support and the named-state bound

For a graph G, write $N_G(X)$ for the open neighborhood of X, $\alpha(G)$ for its independence number, and $\tau(G) = |V(G)| - \alpha(G)$. The following graph statements apply to isolated vertices as well.

Lemma 10 (Near-maximum support)**.** *Suppose every vertex of G lies in an independent set of size at least $\alpha(G) - d$, where $d \geq 0$, and let $t = \tau(G)$. The number N of nonisolated vertices satisfies $N \leq t(d + 2)$ (the support inequality of Gitler–Valencia [3]). If $d > 0$ and $\delta = t(d + 2) - N$, the number of maximum independent sets is at most $2^{\lfloor \delta/d \rfloor}$.*

Proof. Fix a maximum independent set I and put $C = V(G) \setminus I$. For each $c \in C$, choose an independent set of size at least $|I| - d$ containing c, and let X_c be its intersection with C. Then $|N_I(X_c)| \leq |X_c| + d$. For any independent $Y \subseteq C$, maximality of I gives $|N_I(Y)| \geq |Y|$. If A is a union of previously chosen X_c, neighborhood submodularity and independence of $A \cap X_c$ give

$$|N_I(A \cup X_c)| - |A \cup X_c| \leq |N_I(A)| - |A| + d.$$

At most t choices cover C, so $|N_I(C)| \leq t + td$. Every vertex of C is nonisolated, and the nonisolated vertices of I are precisely $N_I(C)$, proving the first assertion.

Every maximum independent set has the unique form $(I \setminus N_I(X)) \cup X$, where $X \subseteq C$ is independent and $|N_I(X)| = |X|$. Let U be the union of these tight exchange sets. The same submodularity argument with $d = 0$ gives $|N_I(U)| - |U| \leq 0$. Add at most $t - |U|$ of the sets X_c to cover C. Thus $N - 2t \leq (t - |U|)d$, or $|U|d \leq \delta$. The maximum sets are indexed by subsets of U. □

Lemma 11 (Independent-neighborhood ratio)**.** *Under the hypotheses of Lemma 10, every independent set I containing no isolated vertex satisfies $|I| \le (d+1)|N_G(I)|$.*

Proof. Suppose $|I| > (d+1)|A|$, where $A = N_G(I)$. If the bipartite graph between A and I has no matching saturating A, choose $Y \subseteq A$ with $|N_I(Y)| < |Y|$. Replace I by $I \setminus N_I(Y)$ and A by its actual neighborhood, a subset of $A \setminus Y$. The strict inequality persists. Repetition terminates with a nonempty violating pair admitting a matching saturating A; a nonempty set of originally nonisolated vertices cannot have empty actual neighborhood.

There is a maximum independent set R containing this I. Indeed, in any maximum set T, replace $T \cap (I \cup A)$ by I. The matching assigns distinct points of $I \setminus T$ to $T \cap A$, so the replacement preserves maximal cardinality and is independent. Put $C = V(G) \setminus R$. For each $a \in A \subseteq C$, choose a near-maximum set containing a and write its intersection with C as $X_a = Y_a \sqcup Z_a$, with $Y_a \subseteq A$ and $Z_a \cap A = \varnothing$. Since Z_a has no neighbors in I, and $|N_R(Z_a)| \ge |Z_a|$,

$$|N_I(Y_a)| + |Z_a| \le |N_R(X_a)| \le |X_a| + d, \qquad |N_I(Y_a)| \le |Y_a| + d.$$

The Y_a cover A. The saturating matching ensures $|N_I(Y)| \ge |Y|$ for every $Y \subseteq A$. Neighborhood submodularity, adding at most $|A|$ sets Y_a, now gives $|I| = |N_I(A)| \le (d+1)|A|$, a contradiction. □

Lemma 12 (Maximum-family intersection and equality)**.** *Let $\mathcal{D}$ be all maximum independent sets of a graph H, all of size s. Put $I_0 = \bigcap \mathcal{D}$, $U_0 = \bigcup \mathcal{D}$, and $B = U_0 \setminus I_0$. Then $|I_0| + |U_0| \ge 2s$. If equality holds, $H[B]$ has a perfect matching, and each member of $\mathcal{D}$ selects exactly one endpoint of each matching edge, together with all of I_0.*

Proof. Fix $R \in \mathcal{D}$. Every member is $(R \setminus N_R(X)) \cup X$ for a tight independent exchange $X \subseteq V(H) \setminus R$, with $|N_R(X)| = |X|$. Let A be the union of the tight exchanges and $D = N_R(A)$. Submodularity as in Lemma 10 gives $|D| \le |A|$. Since $I_0 = R \setminus D$ and $U_0 = R \cup A$, the inequality follows.

In the equality case $|D| = |A|$. Process the tight exchanges successively. Their intermediate union A' has nonpositive neighborhood deficit $|N_R(A')| - |A'|$, which cannot increase; its final value is zero. Consequently every intermediate deficit is zero. When adding X, equality forces

$$N_R(A') \cap N_R(X) = N_R(A' \cap X), \qquad |N_R(A' \cap X)| = |A' \cap X|.$$

Every subset of the independent set X satisfies Hall's inequality, by maximality of R. A matching of X onto $N_R(X)$ therefore exists, and it matches $A' \cap X$ onto its equal-size neighborhood. Its remaining edges extend any already constructed matching of A' onto $N_R(A')$. Induction gives a matching of A onto D. These are the two halves of B. Every maximum set has $|A|$ points in B and at most one on each matching edge, proving the assertion. □

Lemma 13 (Actual-image rank)**.** *In a strongly connected synchronizing automaton, the span V_i of the trajectory rows $\chi_S D_w$ for $|w| \le i$ has dimension at least $\min(i+1, n)$. Consequently, if these images are uncompressed members of a family $\mathcal{F}$, then $\operatorname{rank} \mathcal{F} \ge \min(i+1, n)$.*

Proof. A plateau $V_i = V_{i+1}$ makes the space invariant under every letter. A reset word followed by a path to any state q supplies the row $|S|e_q$, so the eventual span is $\mathbb{R}^Q$. Starting with dimension one, the dimensions strictly increase until n. □

Lemma 14 (Saturated transversal support)**.** *Let $U \subsetneq Q$ be partitioned into $s \ge 2$ nonempty blocks, let $\mathcal{T}$ be their transversal family, and put $k = |U| - s \ge 1$. Suppose every image of an s-set $S \in \mathcal{T}$ through depth k belongs to $\mathcal{T}$, in a strongly connected synchronizing automaton. There is a state $z \in U$ such that every transition from U to its complement has source z. Every transversal avoiding*

z maps to a transversal under every letter. The shortest path from S to z has length k, and z lies in a nonsingleton block.

Write $a = |Q \setminus U|$, $m = k + a$, and let J consist of the singleton blocks. If $c(S) \geq h + m + 1$, then

$$\sum_{v \notin U} |N_{P_h}(v) \cap J| \leq a(a+1)/2.$$

Proof. The reachable unions $U_i = \bigcup_{|w| \leq i} Sw$ grow strictly while proper. Since $|U_k| \leq |U| = s + k$, each of the first k steps adds exactly one state, and $U_k = U$. Write z for the last added state. Every transition from U_{k-1} already lands in U, so every exit from U has source z. Every short image contains every singleton block, hence z is not such a block.

Let W consist of rows supported on U with equal sums on all s blocks. It has dimension $k+1$, and $W_z = W \cap \{x_z = 0\}$ has dimension k. The space V_{k-1} of Lemma 13 lies in W_z and has dimension at least k, hence equals W_z. For any letter f, $V_{k-1} D_f \subseteq V_k \subseteq W$. Thus the row of every transversal avoiding z maps into W. Its nonnegative integer coordinates have total s, so each block sum is one and its image is again a transversal.

For $v \notin U$, let $\rho(v)$ be its distance from z. A shortest such path stays outside U after its first step: returning to U would require another passage through z before exiting, and that cycle could be removed. Thus $\rho(v) \leq a$. For an uncompressed s-set T, put $\varepsilon(T) = \min_{X \in \mathcal{T}} |T \setminus X|$. Changing at most one point in a transversal makes it avoid z. Since function images do not increase one-sided set difference,

$$|Tf| = s \quad \Longrightarrow \quad \varepsilon(Tf) \leq \varepsilon(T) + 1.$$

Reach an actual transversal containing z in k steps and then follow the $\rho(v)$-step path. If compression has not occurred, the resulting image contains v and omits at most $\rho(v)$ points of J. More than $\rho(v)$ neighbors of v in J would therefore produce a P_h edge and compression within $k + \rho(v) + h \leq m + h$, a contradiction. Hence $|N_{P_h}(v) \cap J| \leq \rho(v)$. The positive distance layers are consecutive; on a vertices their sum is at most $1 + \cdots + a$, proving the stated bound. □

Lemma 15 (Finite named support)**.** *Suppose $|S| = n - m \geq 2$, $c(S) \geq h + m + 1$, $h \geq 1$, and $1 \leq \ell \leq m - 1$. Let $G = P_h$, $H = P_{h+\ell}$, $t = \tau(G)$, $d = m - t$, and suppose $\alpha(H) = n - m$. For the complete maximum family $\mathcal{D}$ of H, put $I_0 = \bigcap \mathcal{D}$, $A_0 = Q \setminus \bigcup \mathcal{D}$, $a = |A_0|$, and $W = K_h \cup (Q \setminus I_0)$. Then $a \leq \ell$, $|W| \leq 2m + ad$; if $d \geq 1$ and $\ell \leq 2d$,*

$$|W| \leq 2m + \ell d - 1.$$

Each row omits at most d early isolated states. In particular the named bounds for $(m, h, \ell, t) = (4, 7, 2, 3)$ and $(5, 9, 2, 4)$ are 10 and 11, respectively; the former already follows from $2m + ad \leq 10$.

Proof. All depth-m images are independent in G and cover Q. Every vertex of G therefore belongs to an independent set of size $n - m = \alpha(G) - d$. All depth-$(m - \ell)$ images belong to $\mathcal{D}$; their union has at least $n - \ell$ points, so $a \leq \ell$. Put $B = (\bigcup \mathcal{D}) \setminus I_0$ and $p = |K_h \cap I_0|$. Lemma 12 gives $|B| \leq 2(m - a)$. Every G-neighbor of I_0 belongs to A_0. The independent-neighborhood ratio gives $p \leq a(d+1)$. Thus $|W| = a + |B| + p \leq 2m + ad$.

For the strict bound it remains to consider $a = \ell$ and $|B| = 2(m-a)$, since either strict inequality saves at least one. Lemma 12 provides a perfect matching on B. Partition $U = Q \setminus A_0$ into its matching edges and the singletons I_0. The family $\mathcal{D}$ is contained in the transversals of this partition, whose defect is $k = m - \ell \geq 1$. All depth-k actual images are transversals. Apply Lemma 14: $p \leq a(a+1)/2$ because each nonisolated point of I_0 has a neighbor in A_0. Consequently

$$|W| \leq 2m + \ell(\ell - 1)/2 \leq 2m + \ell d - 1 \quad (1 \leq \ell \leq 2d).$$

Finally the complement of each row is a cover of size m; at least t of its vertices are nonisolated in G. □

Lemma 16 (Early-core filters)**.** *Under $|S| = n - m \geq 2$, $c(S) \geq L$, and $1 \leq h \leq L - m - 1$, put $t = \tau(P_h)$ and $T = \binom{m+2}{2}$. Then*

$$\Delta(P_h) \leq m, \qquad |K_h| \leq t(m - t + 2), \qquad |E(P_h)| \geq h.$$

If $0 \leq b = T + h - L \leq 3m - 2$, then $|E(P_h)| \leq b$. When $t = m$, the maximum-independent-set row rank is at least $\min(L - h, n)$.

Proof. The depth-m images are independent in P_h and cover Q. An independent $(n - m)$-set containing a vertex excludes all its neighbors, giving the degree bound. Lemma 10 with $d = m - t$ gives the support bound. A short-pair layer cannot plateau before becoming complete: its recurrence closes under one-letter preimages and direct kernels, and a plateau would prevent any remaining pair from ever merging. Since S is independent and has at least two points, the first h layers are proper and contribute at least one new edge each.

For the upper edge bound, consider any graph of maximum degree at most m and cover number at most m with fewer than $3m$ edges. It admits an $(n - m)$-set inducing exactly one edge. Indeed, take a minimum cover C of size t' and its independent complement I. If $c \in C$ has $d_I(c) \leq m - t' + 2$, retain one of its neighbors, delete the others from I, and add c, then trim to size $n - m$. If no such c exists, $t' \leq 2$ contradicts the degree bound; for $3 \leq t' \leq m$ there are at least $t'(m - t' + 3) \geq 3m$ edges, again a contradiction. Deleting the one induced edge and repeating gives a Frankl–Pin sequence using every edge: no later edge lies in an earlier chosen set. Thus the maximum sequence length p over any graph obeys $p \geq \min(|E|, 3m - 1)$.

The Frankl–Pin refinement [4] gives $c(S) \leq T - p + h$. For completeness, if a shortest compressing word has length $L' > h$, take its first $L' - h$ prefix images and at each prefix a pair merged by the corresponding suffix. A later selected pair belongs to no earlier image, since otherwise that suffix would shorten the word. None of the images contains a pair merging within h letters. Appending a length-p sequence over P_h gives a Frankl–Pin sequence of length $L' - h + p \leq T$. If $|E(P_h)| \geq b + 1$, this implies $c(S) \leq T - (b + 1) + h = L - 1$. Finally depth-$(L - h - 1)$ images are maximum independent sets when $t = m$, and Lemma 13 gives their rank bound. □

Lemma 17 (Defect-five low-cover core size)**.** *Suppose $n \geq 16$, $|S| = n - 5$, $c(S) \geq 15$, and $\tau(P_9) = 4$. Then $|K_9| \leq 11$. If $|K_9| = 11$, the maximum-independent-set family of P_9 has exactly two members differing by one state; P_{10} retains exactly one of these members, which every letter maps into that same two-member family.*

Proof. Lemma 16 first gives $|K_9| \leq 12$. Suppose $|K_9| \geq 11$. Since S is independent in P_{10}, $4 \leq \tau(P_{10}) \leq 5$. If $\tau(P_{10}) = 5$, apply Lemma 15 with $m = 5$, $h = 9$, $\ell = 1$, $d = 1$ to obtain $|W| \leq 10$, contradicting $K_9 \subseteq W$. Thus $\tau(P_{10}) = 4$. Every maximum independent set of P_{10} maps under every letter, injectively, into a maximum independent set of P_9.

In Lemma 10, the parameters $t = 4$, $d = 1$ give deficit $\delta = 12 - |K_9|$. If $|K_9| = 12$, there is a unique maximum set R of P_9. It is also the unique maximum set of P_{10} and satisfies $Rf = R$ for every letter, contradicting strong connectivity. For $|K_9| = 11$, the proof of that lemma gives a tight-exchange union of size at most one. Hence there are at most two maximum sets, and if two exist they differ by one state. A unique maximum set is excluded as above. Denote the two by R, T. If both remained independent in P_{10}, each letter would map both into $\{R, T\}$ and preserve their proper union of size $n - 3$, again contradicting strong connectivity. Exactly one remains; its identity is common to the entire alphabet. □

Lemma 18 (Low-cover row capacity)**.** *Let $G \subseteq H$ be graphs with $\alpha(G) = s+1$ and $\alpha(H) = s$. Write $\mathcal{M}$ for the maximum independent sets of G, $\mathcal{B}$ for its inclusion-maximal independent sets of size s, and $\mathcal{D}$ for the maximum family of H. Then $|\mathcal{D}| \le |\mathcal{B}| + 2|\mathcal{M}|$. If Z is the union of the G-isolates omitted by members of $\mathcal{D}$, then $|Z| \le \max(2, |\mathcal{M}|)$.*

Proof. A row of $\mathcal{D}$ is either in $\mathcal{B}$ or is obtained by deleting one point from a member R of $\mathcal{M}$. There are at most two such deletions for a fixed R: three deletions would make every pair of R co-occur in an H-independent row, contrary to $\alpha(H) = s$. This proves the cardinality bound, allowing overcounting between maximum sets.

If a row omits an isolated vertex z of G, adding z gives a member of $\mathcal{M}$. If some R supplies two such rows, $R \setminus \{z\}$ and $R \setminus \{z'\}$, then zz' is an edge of H: it is the only pair of R absent from both rows, and some edge must occur in $H[R]$. Every row of $\mathcal{D}$ must omit z or z'. Its complement, a size-$(|V| - s)$ cover of G, can omit at most one G-isolate since $\tau(G) = |V| - s - 1$. Thus $Z \subseteq \{z, z'\}$. Otherwise each member of $\mathcal{M}$ supplies at most one omitted isolate, giving $|Z| \le |\mathcal{M}|$. □

Lemma 19 (Necessary letter signatures)**.** *Let $\mathcal{D}$ be the complete later maximum-independent-set family described above. For each actual letter f, record its action on the named support, the common-reservoir entrance and exit counts, and the images of all rows in $\mathcal{D}$. Each signature satisfies the following finite conditions:*

1. *Every safe row image has the prescribed cardinality, and its named part lies in a maximum or near-maximum independent set of the early graph.*
2. *Every pair in the early graph has a merging word of the designated length, and the one-step pair-distance recurrence holds for the candidate letter pool.*
3. *For the selected alphabet as a whole, the union of its named-state and reservoir arrows is strongly connected. This condition is imposed on the joint selection, not on an individual letter.*
4. *When $H = h + 2$, each letter square maps every later safe row to an early independent set, with the same named-state identity matching in its two applications.*

If the complete finite pool admits no compatible joint row and pair distances, then the associated long-compression hypothesis is excluded for every finite alphabet.

Proof. The first condition follows from independence in P_h. The second is the defining shortest-merge recurrence for P_h. Every actual path projects to a path in the finite named-state/reservoir graph, giving the third. The fourth follows from $H - h = 2$: a compressed image or an early pair collision after the two-letter prefix would produce compression before the proposed threshold. The identical letter has the identical action at both positions. Every letter of an actual automaton supplies a signature satisfying all four conditions. A compatible actual alphabet therefore supplies a jointly feasible selection from the complete signature pool. Exhaustion of the latter excludes the former. □

Lemma 20 (Seven-layer reduction)**.** *Under $n \ge 11$, $|S| = n-4$ and $c(S) \ge 12$, the branch $\tau(P_7) = 4$ is confined to nonisolated cores with $5 \le |K_7| \le 8$, degree at most four and $7 \le |E(P_7)| \le 10$. Their complete isomorphism classification comprises 91 cores.*

Proof. Apply Lemma 16 with $(m, L, h) = (4, 12, 7)$. It gives degree at most four, support at most eight, seven through ten edges and row rank at least five. Cover number four forces at least five

nonisolated vertices. Complete isomorphism enumeration with these filters gives $1, 17, 47, 26$ cores at orders $5, 6, 7, 8$, respectively. Section B specifies the complete finite-domain generation and necessary joint-distance test. □

Lemma 21 (Nine-layer reduction)**.** *Under $n \geq 16$, $|S| = n - 5$ and $c(S) \geq 15$,*

$$\tau(P_9) \in \{4, 5\}.$$

The cover-five branch has nonisolated support between six and ten states, at least nine early edges, maximum degree at most five and maximum-family row rank at least six. The complete classification retains 2,294 core graphs.

Proof. The set S is independent in P_9, giving cover number at most five. Lemma 9 applied to all $(n-3)$-sets gives cover number at least four. In the cover-five case, depth-five actual images are maximum independent sets and cover all states. Lemma 16, with $(m, L, h) = (5, 15, 9)$, gives the stated degree, support, edge and rank filters. Complete generation of graphs of orders six to ten with degrees one through five, followed by the precise cover, edge and row tests, yields 2,294 isomorphism classes. The corresponding joint one-letter and distance domains are handled in Section 4. □

Lemma 22 (Auxiliary maximum source)**.** *Suppose $n \geq 16$, $c(S) \geq 15$, $\tau(P_9) = 4$ and the nonisolated support of P_9 contains eleven vertices. Then $\tau(P_{10}) = 4$. The maximum-independent-set family of P_9 consists of two sets R, T differing at one state. Exactly one member $R_0 \in \{R, T\}$ remains independent in P_{10}, and every actual letter maps this same R_0 into $\{R, T\}$.*

Proof. This is the eleven-vertex case of Lemma 17. In particular, its source is selected by P_{10}, once for the entire alphabet, rather than separately for individual letters. □

Lemma 23 (Core entrance bound)**.** *Let $G = P_h$ in a synchronizing automaton, with $h \geq 2$ and proper nonisolated support $V \subsetneq Q$. If a letter f sends an isolated state into $x \in V$, then*

$$\deg_G(x) \leq d(f) + F_G(f) \leq |E(G)| - h + 2,$$

where $F_G(f)$ counts distinct edges in G having a G-nonedge preimage under f.

Proof. For each neighbor of x in the image Qf, choose a preimage. Together with the isolated preimage of x, it gives a G-nonedge, and its image is a G-edge. At most $d(f)$ neighbors lie outside Qf. The nontrivial fibers of f give at least $d(f)$ distinct kernel pairs in P_1. Every edge counted by $F_G(f)$ has merging distance exactly h: a shorter merge, preceded by f, would merge its nonedge preimage within h letters. The layers $P_i \setminus P_{i-1}$, $2 \leq i \leq h-1$, are nonempty. Indeed, a pair of merging distance greater than h exists because G is proper; the suffixes of a shortest merging word supply every intermediate distance. These disjoint layers contain at least $d(f) + (h-2) + F_G(f)$ edges in total. □

Lemma 24 (Proper-layer pair entrance)**.** *Let P_h be a proper short-pair layer of a synchronizing automaton, $h \geq 1$. Some actual letter f sends a pair outside P_h to an edge of P_h.*

Proof. Choose a pair of states with minimal merging distance $d > h$ and follow a shortest merging word. Its successive pair distances decrease by at most one, and the first step from distance $h+1$ reaches distance h. The corresponding original letter provides the required entrance. □

B Finite-domain generation and joint feasibility

The finite calculation uses the support and rank bounds of Appendix A to specify its input domain. Within these bounds it enumerates graphs, closed row families and necessary one-letter signatures. A joint feasibility test then asks whether some collection of signatures supports the required distances.

B.1 Graph and row-family coverage

For the defect-five cover-five branch, the graph6 input contains one representative of each graph of orders $6, \dots, 10$ with minimum degree at least one and maximum degree at most five. The driver filters by cover number, edge count, union of the maximum independent sets and row rank. The input includes disconnected graphs. The cover-four classification instead fixes a four-vertex cover C. Its complement is independent, so its vertices are represented by nonempty neighborhoods in C. The generator enumerates multisets of these neighborhoods and all $2^{\binom{4}{2}} = 64$ possible edge sets inside C, then applies the stated filters and graph-isomorphism reduction. Every eligible graph has such a representation: choose a minimum cover, label it by $0, 1, 2, 3$, and order the remaining vertices by their neighborhoods. This proves completeness of the presentations before isomorphism reduction.

For a fixed early core K, name all exceptional isolated vertices E that are absent from at least one later row. The remaining isolated reservoir J belongs to every row. Each candidate later row is either an early maximum independent set with one vertex deleted, or a nonextendable near-maximum independent set, with the appropriate isolated vertices included. These are all possibilities because a later row is early-independent and has size one below the early independence number.

Closure uses the pairs that co-occur in selected rows, including diagonal pairs to record whether a vertex occurs at all. For a candidate family $\mathcal{F}$, let $U(\mathcal{F})$ be this set of pairs and define

$$\mathrm{cl}(\mathcal{F}) = \{R : R \text{ is a candidate row and every pair of } R \text{ belongs to } U(\mathcal{F})\}.$$

A complete maximum-independent-set family is closed: any candidate whose pairs all occur in independent rows is itself independent. A selection is rejected if the same pair data makes an early maximum set independent in the later layer. The implementation stores the pair data in 128 bits, sufficient for the 66 diagonal and off-diagonal positions on eleven named vertices.

The recursive enumeration adds a row, closes the selection, and rejects a branch if closure introduces a previously skipped earlier row. For any target closed family, its first missing row specifies the next branch; closure remains inside that target. Thus every closed family is reached, and the earlier-row rule gives it a unique construction path. Rank, union and cardinality filters are applied with their proved bounds. Core automorphisms and a single consistent renaming of exceptional isolates remove equivalent representations. The common reservoir is kept separate from the named exceptional isolates throughout this construction.

The rank computation uses the field $\mathbb{F}_{1009}$ and agrees with rational rank on these row matrices. Every zero–one minor of order at most ten has absolute determinant below 522, by the bordered Hadamard bound $(k+1)^{(k+1)/2}/2^k$. At eleven named vertices, the only additional case is an 11×11 minor of later rows, each of weight six. Its determinant is divisible by six: replace one column by the sum of all columns and factor out six. The same bound is 1458, below $6 \cdot 1009$. Thus a nonzero integer minor cannot vanish modulo 1009. This also applies to every subfamily used in the safe-tail rank constraints.

B.2 One-letter signatures and the SAT implication

A safe letter is injective on every later row. Its kernel on the named support therefore consists of blocks compatible with the early pair graph. The generator enumerates these partitions and the possible incoming images of common isolated states. For a fixed kernel and incoming set, it enumerates injections of the ordered kernel blocks into available output labels. The independent generation order enumerates the same finite maps in a different order; comparison uses exact permutation ranks. Cardinalities, row images and pair images are retained in the signature. Common-reservoir input and output labels are distinguished; a square test identifies them consistently when applying the same letter twice.

Let d_R be the exit distance of a retained row and d_p the merging distance of a retained pair. Every chosen signature obeys the one-step inequalities

$$d_x \le 1 + d_{xf},$$

with terminal distance zero. Each nonterminal row or pair has at least one chosen signature attaining equality. These are Bellman conditions for shortest distance. The encoding additionally enforces the safe-tail rank bounds and the required slow row. Strong connectivity is imposed on the *union* of the named-state and reservoir arrows of the compatible alphabet, in both directions. In the eleven-point branch the auxiliary source R_0 is fixed before generating the whole alphabet; the two possible choices are separate branches. In the strengthened predicate, an existential clause requires a chosen signature supplying the proper-layer entrance of Lemma 24.

An actual automaton supplies its letter signatures and true shortest distances, and hence satisfies every constraint above. Consequently UNSAT excludes an actual alphabet in the specified domain. A feasible assignment records a necessary relaxation; the paper overlap lemma resolves the historical exceptional assignment.

B.3 Reported coverage and reproduction

The historical completion records summarized in Section 4 use the following partition:

branch	canonical necessary domains
$m = 4,\ \tau(P_7) = 4$	91 cores
$m = 4,\ \tau(P_7) = 3$	10 142
$m = 5,\ \tau(P_9) = 5$	2 294 cores
$m = 5,\ \tau(P_9) = 4,\ N = 11$	3 595
$m = 5,\ \tau(P_9) = 4,\ 5 \le N \le 10$	3 206 988

For cover five, 71 intervals of length 32 and a final interval of length 22 cover the retained-core order. The defect-four low-cover partition has six intervals of sizes $1414, 259, 978, 2349, 2500, 2642$. The eleven-point branch combines $[883 + (1201 - 19)] + [1504 + 26] = 3595$ domains, counting the 19-domain overlap once. For early support five to ten, $3\,245\,386 - 38\,398 = 3\,206\,988$. The original predicate yielded 3 206 987 UNSAT decisions and the single $K_{4,4}$ relaxation treated in Lemma 5.

The source revision in [8] provides executable commands. Its explicit option

```
-historical-no-pair-entrance
```

used with `--distances` and `--joint-sat`, disables the later entrance clause. This reconstructs the older predicate within the current implementation. Each full reproduction records the source and solver versions, graph6 input, all canonical interval boundaries, selected and completed domain

counts, feasibility total, exit status and terminal completion summary. Static classification counts and dynamic completion counts are recorded separately.

The present revision's checks comprise the bounded joint-SAT control, the complete target-capacity control, the bounded literal graph/closed-family control, and the selected $K_{4,4}$ comparison. The entrance-enabled joint-SAT control checked 100,686 instances, including 3,570 feasible instances. The capacity control examined 4,390,434 incoming sets over 4,658 cores, with at most 42 admissible targets. The graph control compared all 33,792 labelled graphs of orders five and six, retaining 30 classes, and checked their closed families with up to two added isolates against literal subset selection. Both selected-domain runs generated 1,768 signatures at core 447, domain 0; the historical mode returned `joint=1`, and the entrance-enabled default returned `joint=0`. The large-domain totals above retain the provenance of the historical completed calculations; these bounded and selected checks constitute the checks performed for this revision.

C Constructions for the smaller values of q

Theorem 25. *The following bounds hold, and every lower bound has a binary witness.*

1. $\mathcal{F}(n, r+1, r) = n - r$ *for* $n \geq 2$ *and* $1 \leq r \leq n-1$.
2. $\mathcal{F}(n, r+2, r) = n - r$ *for* $n \geq 4$ *and* $1 \leq r \leq n-3$.
3. *For* $r \geq 1$, $s - r \geq 3$ *and* $n \geq 2s - r - 1$, *one has*

$$\mathcal{F}(n, s, r) = n - r.$$

For $1 \leq r < n$, *one has* $\mathcal{F}(n, n, r) = 1$.

Thus the first two cardinality faces are determined, including their full-image boundary. For every fixed (s, r) with $s - r \geq 3$, the bound is sharp at every $n \geq 2s - r - 1$, in particular at $n = r + k(s - r - 1) + 2$ for $k \geq 2$.

We recall the upper bound with a proof, to make the argument self-contained. For a word w let $D_w(q, t) = 1$ if $qw = t$, and 0 otherwise. Then $D_{uv} = D_u D_v$. The characteristic vector χ_S is a row vector; $\chi_S D_w$ counts trajectories from S, with multiplicity, and its support is Sw.

Lemma 26 (Szykuła [6, 7])**.** *If* $A \not\subseteq Sz$ *for some word* z, *then* $\mathrm{ac}(S, A) \leq n - |A|$.

Proof. Write $r = |A|$, $s = |S|$, and let V_i be the span over $\mathbb{R}$ of $\chi_S D_w$ for $|w| \leq i$. Suppose all words of length at most $n - r$ are safe. Safety gives $|Sw| = |S|$, so $S \to Sw$ is injective: no trajectories collide, and every image state has multiplicity 1. Their rows therefore have total s and value 1 at every coordinate of A. Hence V_{n-r} lies in

$$H = \{x : x_q = s^{-1} \textstyle\sum_{t \in Q} x_t \text{ for every } q \in A\}.$$

The r equations are independent since $r < s$, so $\dim H = n - r$. Now $\dim V_0 = 1$, and $V_i = V_{i+1}$ implies invariance under every letter and stabilization forever. Therefore some equality must occur before index $n - r$, or $\dim V_{n-r} \geq n - r + 1$. Stabilization puts $\chi_S D_z$ in H, contradicting its zero coordinate in A and total s. □

In a strongly connected synchronizing automaton with $n \geq 2$, a reset word followed by a path to a state outside A gives the required z. This proves every upper bound $n - r$ in Theorem 25. If

$S = Q$, some letter must be nonpermutational and immediately compresses Q, while the empty word is safe. Thus $\mathrm{ac}(Q, A) = 1$.

We will also use the following elementary fact. If t cycles $x_0, x_1, \ldots, x_{s-1}$ and e sends x_0 to x_1 while fixing the other points, then

$$e(t^{s-1}e)^{s-2} \tag{4}$$

maps this set to x_1. Indeed, after e the image is $\{x_1, \ldots, x_{s-1}\}$; each $t^{s-1}e$ deletes the last point of the remaining interval. This also covers $s = 2$.

C.1 The first two cardinality faces

Proposition 27. *Part (1) of Theorem 25 holds.*

Proof. Put $L = n - r \geq 1$ and $Q = \{0, \ldots, n-1\}$. Define

$$qa = \begin{cases} q+1 & q < L, \\ q & q \geq L, \end{cases} \qquad S = \{0, L, L+1, \ldots, n-1\}, \qquad A = S \setminus \{0\}.$$

Let b cycle S in the displayed order and send every point outside S to L. Then $Qb = S$. The a-path from 0 to L and the b-cycle prove strong connectivity.

For $0 \leq j < L$, set $S_j = A \cup \{j\}$. One has $S_j b = S_0$, and $S_j a = S_{j+1}$ for $j < L-1$; $S_{L-1}a = A$. Induction shows that every word of length less than L has image S_j with $j < L$, so is safe. The word a^L compresses S, proving $\mathrm{ac}(S, A) = L$.

On S, $c = a^L$ is the unit $0 \mapsto L$. Formula (4) with $t = b$ shows that $c(b^r c)^{r-1}$ resets S. It resets Q as well, since $Qc = A = Sc$. This proves synchronization, also when $L = 1$ or $r = 1$. □

Proposition 28. *Part (2) of Theorem 25 holds.*

Proof. Put $M = n - r \geq 3$. Take disjoint sets $A = \{p_0, \ldots, p_{r-1}\}$ and $U = \{u_1, \ldots, u_M\}$, and let $S = A \cup \{u_1, u_2\}$. Letter a fixes A, while

$$p_i b = p_{i+1} \quad (0 \leq i < r-1), \qquad p_{r-1} b = u_1.$$

For odd M, let a cycle $u_1, \ldots, u_M$ and set

$$u_j b = \begin{cases} p_0 & j \text{ odd}, \\ u_2 & j \text{ even}. \end{cases}$$

For even M (necessarily $M \geq 4$), set

$$u_1 a = u_2, \quad u_j a = u_{j+1}\ (2 \leq j < M), \quad u_M a = u_2, \qquad u_j b = \begin{cases} u_2 & j = 1 \text{ or } j \geq 3 \text{ odd}, \\ p_0 & j \text{ even}. \end{cases}$$

In both cases $Qb = S$. From p_0 the b-path reaches u_1, whose a-orbit covers U; some state of U returns to p_0 under b. Every state therefore reaches and is reached from p_0.

The two moving points of Sa^j remain distinct for all $j \geq 0$. For $0 \leq j \leq M-2$, their b-images are p_0 and u_2, so $Sa^j b = S$. At $j = M-1$, the pair is $\{u_M, u_1\}$ for odd M and $\{u_M, u_2\}$ for even M; both points map to p_0. Thus $h = a^{M-1}b$ compresses S. Every word shorter than M splits into a-runs followed by b, each run having length at most $M-2$, and a trailing a-run. All these transitions are safe. Hence $\mathrm{ac}(S, A) = M$.

Set $s = r + 2$, and $t = ab$ for odd M, $t = b$ for even M. On S, t cycles $p_0, \dots, p_{r-1}, u_1, u_2$. If e is the unit $u_1 \mapsto u_2$ on S, then $h = et$ there. Consequently

$$h(t^{s-2}h)^{s-2} = e(t^{s-1}e)^{s-2}t$$

on S, and this is a reset by (4). Prepending b gives a reset of Q. The unit e is used only to verify the identity; the reset word uses the original letters. □

C.2 Every larger fixed gap

Proposition 29. *For $r \geq 1$, $g \geq 2$ and $M \geq 2g$, there is a binary strongly connected synchronizing automaton with*

$$n = r + M + 1, \quad |S| = s = r + g + 1, \quad |A| = r, \qquad \mathrm{ac}(S, A) = M + 1.$$

The subset S is an actual reachable image.

Proof. Take disjoint sets $P = \{p_0, \dots, p_r\}$ and $U = \{u_1, \dots, u_M\}$. Set $A = P \setminus \{p_r\}$ and $S = P \cup \{u_1, \dots, u_g\}$. First suppose $M \geq 2g + 1$ and put $h = 1 + ((M - 1) \bmod g)$. Then $1 \leq h \leq g$ and $M \geq 2g + h$. Let a fix P, send u_j to u_{j+1} for $j < M$, and send u_M to u_{h+1}. Define $p_0 b = u_h$, $p_i b = p_{i-1}$ for $1 \leq i \leq r$, and

$$u_j b = \begin{cases} p_r & j \equiv h - 1 \pmod g, \\ u_1 & j = g,\ h \geq 2, \\ u_{h+g} & j = M, \\ u_{j+1} & \text{otherwise.} \end{cases}$$

The cases are disjoint. On S, b cycles $p_0, u_h, \dots, u_g, u_1, \dots, u_{h-1}, p_r, \dots, p_1$. The segment $u_1, \dots, u_{h-1}$ is empty for $h = 1$. Ordinary outside states follow increasing indices to the next index congruent to $h - 1$, then enter p_r. The exceptional path of length $g + 1$ is

$$u_M \longrightarrow u_{h+g} \longrightarrow \cdots \longrightarrow u_{h+2g-1} \longrightarrow p_r.$$

Its indices exist because $M \geq 2g + h$. Thus $Qb^{g+1} = S$. The a-path from u_1 covers U, and the b-cycle covers P, proving strong connectivity.

Safety. Use singleton P-cells and color cells $1, \dots, g$, assigning u_j color $1 + ((j - h - 1) \bmod g)$. Let C be the $n \times s$ state-to-cell incidence matrix. Let T_a fix P and cycle the colors, and let T_b cycle $p_0, g, 1, \dots, g - 1, p_r, \dots, p_1$. For a standard state column f_q and cell row e_c, one has

$$D_a C = C T_a, \qquad D_b C = C T_b + f_{u_M}(e_g - e_1). \tag{5}$$

The last term is an outer product of an $n \times 1$ column and a $1 \times s$ row. These cells are bookkeeping devices, not a quotient automaton. For $w = \ell_1 \cdots \ell_t$, iteration gives

$$\chi_S D_w C = \mathbf{1}_s + \sum_{\substack{1 \leq i \leq t \\ \ell_i = b}} \mu_i (e_g - e_1) T_{\ell_{i+1} \cdots \ell_t}, \qquad \mu_i = (\chi_S D_{\ell_1 \cdots \ell_{i-1}})_{u_M} \geq 0. \tag{6}$$

This counts trajectories even after collisions, with products in word order. A trajectory from S needs at least $M - g$ letters to reach u_M: initial indices are at most g, every transition within U increases an index by at most one, and entrances from P have index $h \leq g$. Hence a nonzero

correction has $i \geq M - g + 1$. In the cell graph, color g needs at least $g - 1$ letters to reach color $g - 1$, and color 1 needs at least g letters to reach A. The latter route traverses $1, \ldots, g - 1$ and then takes two letters b through p_r. A detour through P cannot shorten either route.

For $t \leq M - 1$, correction suffixes have length at most $g - 2$, so no positive correction reaches color $g - 1$, whose trajectory count is at most one. Suppose a word of length at most M has a first collision. For a colliding b, the distinct source states must lie in its only non-singleton fiber, all of color $g - 1$, contradicting this count bound. For a colliding a, the sources must be u_h, u_M, its only non-singleton fiber. They are not both initially present. A preceding b cannot give u_M, which has no b-preimage. A preceding a requires sources u_{h-1}, u_{M-1} one step earlier, both of color $g - 1$, again a contradiction; for $h = 1$, u_h has no a-preimage at all. Thus every word of length at most M preserves s states. For $t \leq M$, every correction suffix has length at most $g - 1$, so no negative correction reaches A. Its singleton cells retain multiplicity at least one. Every such word is safe.

The word $a^{M-g}b^{g+1}$ has length $M+1$. Exactly its first b has a trajectory at u_M, with multiplicity one. Its negative correction is $e_1 T_b^g = e_{p_{r-1}}$, whereas its positive correction is $e_g T_b^g = e_{p_r}$. It therefore omits p_{r-1} and attains equality, also when $r = 1$.

For synchronization put $c = a^{M-h}b^s$. The a-run fixes the cyclic tail $u_{h+1}, \ldots, u_M$. For $j < h$, it sends u_j to u_{M-h+j}, which has the same remaining b-distance $h - j$ to p_r as u_j. For $j = h$, it sends u_h to u_M, which reaches p_0 in $g + 1 + r = s$ letters b. Consequently c fixes $S \setminus \{u_h\}$ and sends u_h to its b-predecessor p_0. After c, each bc extends the missing interval by one point in forward b-order. Thus $b^{g+1}c(bc)^{s-2}$ resets Q.

The boundary $M = 2g$. Let a fix P and cycle U. Define $p_0 b = u_g$, $p_i b = p_{i-1}$ and

$$u_j b = \begin{cases} p_r & j \equiv g - 1 \pmod{g}, \\ u_{j+1} & j < M,\ j \not\equiv g - 1 \pmod{g}, \\ u_M & j = M. \end{cases}$$

Now b cycles $B = P \cup \{u_g, \ldots, u_{2g-1}\}$ and fixes u_{2g}; $Qb^{g+1} = E = B \cup \{u_{2g}\}$. Put $d = a^{g-1}b^s$. It fixes P and on E has actions

$$u_g d = u_{2g-1}, \quad u_{g+1} d = u_{2g}, \quad u_{g+j} d = u_{g+j-1}\ (2 \leq j \leq g - 1), \quad u_{2g} d = u_{2g-1}.$$

Therefore $Eda^g = S$, proving actual reachability. The two cycles prove strong connectivity. Assigning u_j color $1 + ((j - 1) \bmod g)$ gives exactly (5)–(6) again. The same delay and collision arguments apply, with a now a permutation. The word $a^g b^{g+1}$ attains avoidance: only the first negative correction has a suffix of g letters b and reaches p_{r-1}; no positive correction reaches A. Finally $e = d^g$ fixes $E \setminus \{u_g\}$ and sends u_g to u_{2g}. Each be removes one further point of B while retaining u_{2g}. Hence $b^{g+1}e(be)^{s-1}$ resets Q. □

Taking $g = s - r - 1$ and $M = n - r - 1$ proves part (3) of Theorem 25.

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